Savannah River National Laboratory®

A U.S. DEPARTMENT OF ENERGY NATIONAL LAB • SAVANNAH RIVER SITE • AIKEN, SC • USA

Determination of Reportable Radionuclides for Defense Waste Processing Facility (DWPF) Sludge Batch 10 (Macrobatch 12)

S. C. Hunter

D. P. Diprete

October 2024

SRNL-STI-2024-00240, Revision 0

srnl.gov

DISCLAIMER

This work was prepared under an agreement with and funded by the U.S. Government. Neither the U.S. Government or its employees, nor any of its contractors, subcontractors or their employees, makes any express or implied:

1. warranty or assumes any legal liability for the accuracy, completeness, or for the use or results of such use of any information, product, or process disclosed; or
2. representation that such use or results of such use would not infringe privately owned rights; or
3. endorsement or recommendation of any specifically identified commercial product, process, or service.

Any views and opinions of authors expressed in this work do not necessarily state or reflect those of the United States Government, or its contractors, or subcontractors.

Printed in the United States of America

Prepared for
U.S. Department of Energy

SRNL-STI-2024-00240
Revision 0

Keywords: *WAPS, DWPF, Sludge Batch, Tank 40*

Retention: *Permanent*

Determination of Reportable Radionuclides for Defense Waste Processing Facility (DWPF) Sludge Batch 10 (Macrobatch 12)

S. C. Hunter
D. P. Diprete

October 2024

Savannah River National Laboratory is operated by Battelle Savannah River Alliance for the U.S. Department of Energy under Contract No. 89303321CEM000080.

REVIEWS AND APPROVALS

AUTHORS:

SETH HUNTER (Affiliate) Digitally signed by SETH HUNTER (Affiliate)
Date: 2024.10.10 14:23:36 -04'00'

S. C. Hunter, Chemical Flowsheet Development Date

DAVID DIPRETE (Affiliate) Digitally signed by DAVID DIPRETE (Affiliate)
Date: 2024.10.11 16:09:52 -04'00'

D. P. Diprete, Environmental and Legacy Management Date

TECHNICAL REVIEW:

CHRISTOPHER MARTINO (Affiliate) Digitally signed by CHRISTOPHER MARTINO (Affiliate)
Date: 2024.10.14 07:29:28 -04'00'

C. J. Martino, Advanced Materials and Process Technology, Reviewed per E7 2.60 Date

APPROVAL:

BRANDI CLARK (Affiliate) Digitally signed by BRANDI CLARK (Affiliate)
Date: 2024.10.14 17:15:52 -04'00'

B. N. Clark, Manager Date
Chemical Flowsheet Development

FRANK PENNEBAKER (Affiliate) Digitally signed by FRANK PENNEBAKER (Affiliate)
Date: 2024.10.15 07:25:49 -04'00'

F. M. Pennebaker, Director, Nuclear and Chemical Processing Date

Robert Hoeppel Digitally signed by Robert Hoeppel
Date: 2024.11.04 18:29:07 -05'00'

R. M. Hoeppel, Manager, DWPF Facility Engineering Date

ACKNOWLEDGEMENTS

The authors would like to acknowledge the Nuclear Measurements group for all radionuclide measurements and Leon Dyers for assistance with all sample digestions. They would also like to thank Shielded Cells management and technicians Forrest Probst, Dee Wheeler, and Michael George for help with sample preparation in the Shielded Cells.

EXECUTIVE SUMMARY

Savannah River National Laboratory (SRNL) was tasked with the radionuclide characterization of the Sludge Batch 10 (SB10) Tank 40 sample (HTF-40-23-24) in accordance with requirements for reporting the Waste Acceptance Product Specifications (WAPS). The Defense Waste Processing Facility (DWPF) is required to report all radionuclides with half-lives greater than ten years and which comprise greater than 0.05% of the total activity inventory for a given waste form at certain specified "index years". DWPF complies with the requirements by considering the half-life requirement ($t_{1/2}$ > 10 years) and radionuclides with concentrations greater than 0.01% of the total inventory from the approximate time of production through 1,100 years.

The following twenty-six radionuclides were identified as reportable in the SB10 Tank 40 sample based on the criteria outlined above. Additional radionuclides may be reportable from the salt streams added during the SB processing at DWPF. All the reportable radionuclides for the SB10 Tank 40 sample are radionuclides that have been reportable in two or more previous sludge batches.

Cl-36	Ni-59	Ni-63	Sr-90	Zr-93
Nb-93m	Tc-99	Cd-113m	Sn-126	I-129
Cs-137	Sm-151	Th-229	U-233	U-234
Np-237	U-238	Pu-238	Pu-239	Pu-240
Pu-241	Am-241	Pu-242	Am-243	Cm-244
Cf-251				

Additionally, the isotope distribution for uranium and plutonium are reported herein to satisfy Specification 1.6 of the WAPS, International Atomic Energy Agency (IAEA) Safeguards. U-235 and U-236 are also reportable in order to comply with the IAEA safeguards.

TABLE OF CONTENTS

LIST OF TABLES ix
LIST OF FIGURES ix
LIST OF ABBREVIATIONS x
1.0 Introduction 1
1.1 Radionuclides Considered 1
2.0 Experimental Procedure 1
2.1 Direct Methods 1
2.1.1 Inductively Coupled Plasma - Mass Spectrometry (ICP-MS) 1
2.1.2 Gamma Counting 2
2.2 Separation Methods 2
2.2.1 Cl-36 Method 2
2.2.2 Ni-59/Ni-63 Method 2
2.2.3 Gamma Counting Following Cs Removal 2
2.2.4 Se-79 Method 3
2.2.5 Sr-90 Method 3
2.2.6 Pd-107 Method 4
2.2.7 Sn-121m method 4
2.2.8 I-129 Method 4
2.2.9 Pm-147/Sm-151 Method 4
2.2.10 Ra-226 Method 4
2.2.11 Th-229/-230, Ac-227 Method 5
2.2.12 Pa-231 Method 5
2.2.13 U-233, U-234, U-235, U-236 Method 5
2.2.14 Pu-238/-241 Method 5
2.2.15 Pu-242/Pu-244 Method 6
2.2.16 Am/Cm Method 6
2.3 Calculated Radionuclide Activities 6
2.3.1 Nb-93m 6
2.3.2 Cd-113m 7
2.3.3 Cs-135 8
2.3.4 Pb-210 and Ra-226 8
2.4 Assumptions 9
2.5 Quality Assurance 9

3.0 Results and Discussion 10
3.1 Summary of Radionuclide Activities for Input 10
3.2 Additional Requested Radionuclides 12
3.3 Determination of Reportable Radionuclides 12
3.4 The Ratio by Weight of U and Pu Isotopes 14
4.0 Conclusions 15
5.0 References 16
Appendix A . Individual Radionuclide Concentrations A-1
Appendix B . Reportable Radionuclides for Sludge Batches 1b to 10. Red Shaded Cells Indicate Reportable Radionuclide. B-1
Appendix C . RadDecay® v.7.00 Output Activities for the Specified Index Years C-1

LIST OF TABLES

Table 2-1. SB10 Calculated FYSF Values 8

Table 3-1. SB10 Radionuclides and Activities Used as Inputs in RadDecay® v. 7.00 11

Table 3-2. List of Additional Radionuclides and Activities 12

Table 3-3. DWPF SB10 Reportability Matrix for 2023 to 3115 – Blue Shaded Cell Indicates The Radionuclide Is Reportable For That Year. 14

Table 3-4. DWPF SB10 Uranium Isotopes Distribution 15

Table 3-5. DWPF SB10 Plutonium Isotopes Distribution 15

LIST OF FIGURES

Figure 3-1. Total and Reportable Radionuclides Activity of DWPF SB10 Tank 40 Sample, as Measured for 2023 and Predicted for Index Years Through 3115 by RadDecay® v. 7.00. The Discrepancy Between the Total and Reportable Activity is Attributable to Radionuclides with Half-lives of Less than 10 Years 13

LIST OF ABBREVIATIONS

ATFC	Analytical and Tank Farm Characterization
CMPO	Octyl phenyl-N,N-di-isobutyl carbamoylphosphine oxide
CST	Crystalline Silico-Titanate
DAAP	Diamyl, amylphosphonate
DMG	Dimethylglyoxime
DOE	Department of Energy
DWPF	Defense Waste Processing Facility
FYSF	Fission Yield Scaling Factor
HLW	High Level Waste
IAEA	International Atomic Energy Agency
ICP-AES	Inductively Coupled Plasma-Atomic Emission Spectroscopy
ICP-MS	Inductively Coupled Plasma-Mass Spectrometry
MST	Monosodium Titanate
NM	Nuclear Measurements
PHA	Pulse Height Analysis
PIPS	Passivated, Implanted, Planar Silicon
RSD	Relative Standard Deviation
SB10	Sludge Batch 10
SRMC	Savannah River Mission Completion
SRNL	Savannah River National Laboratory
SRS	Savannah River Site
TBP	Tri-n-butylphosphate
TTA	2-Thenoyltrifluoroacetone
TTQAP	Task Technical and Quality Assurance Plan
TTR	Technical Task Request
WAPS	Waste Acceptance Product Specifications
WCP	Waste Form Compliance Plan
WQR	Waste Form Qualification Report

1.0 Introduction

A 3-L Tank 40 sample (HTF-40-23-24) was collected on February 28, 2023, by Savannah River Mission Completion (SRMC) following a transfer from Tank 51 to achieve the final Sludge Batch 10 (SB10) processing composition. Savannah River National Laboratory (SRNL) was tasked with characterization of the sample in accordance with requirements for reporting the Waste Acceptance Product Specifications (WAPS). The SB10 WAPS sample was also analyzed for chemical and physical compositions and the results were reported previously.[1]

In accordance with the Department of Energy (DOE) WAPS,[2] DWPF is required to report all radionuclides with half-lives greater than ten years and which comprise greater than 0.05% of the total activity inventory for a given waste form at certain specified "index years." DWPF complies with the requirements by considering the half-life requirement ($t_{1/2}$ > 10 years) and radionuclides with concentrations greater than 0.01% of the total inventory from the approximate time of production through 1100 years. In past WAPS analyses the index years were set at 2015 and 3115; however, starting with Sludge Batch 9, the initial index year will be the year of radionuclide analysis and the final index year will remain at 3115.[3] The initial set of radionuclides to be listed was based on the DWPF Waste Form Compliance Plan (WCP)[4] and the Waste Form Qualification Report (WQR)[5] and is expanded upon when additional radionuclides meet the reportability criterion for a given index year.

Specification 1.6 of the WAPS, International Atomic Energy Agency (IAEA) Safeguards Reporting for High Level Waste (HLW), requires that the ratio by weights of the following uranium and plutonium isotopes be reported: U-233, U-234, U-235, U-236, U-238, Pu-238, Pu-239, Pu-240, Pu-241, and Pu-242. As such, a complete list of the Pu and U radionuclides measured during SB10 radionuclide analysis is provided in this report.

These analyses were requested by DWPF via Technical Task Request (TTR) X-TTR-S-00075, Rev. 3; the SRNL activities that satisfy the request are outlined and directed by a Task Technical and Quality Assurance Plan (TTQAP), which included an analytical study plan for the analyses.[6,7]

1.1 Radionuclides Considered

Previous sludge batch reports have listed radionuclides that may be reportable.[8,9] The lists were based on knowledge of the various Savannah River Site (SRS) processes that generated the waste. No new radionuclides have been added to the list for the SB10 analyses.

2.0 Experimental Procedure

The preparation of the SB10 WAPS sample used in the radionuclide analysis was described in a previous document.[1] The following radionuclide method descriptions and calculations have been given previously, most recently for SB8 and SB9.[9,10]

2.1 Direct Methods

2.1.1 Inductively Coupled Plasma - Mass Spectrometry (ICP-MS)

Inductively coupled plasma – mass spectrometry (ICP-MS) was employed to analyze separate subsamples of the aqua regia digestions of the Tank 40 SB10 dried solids described in a previous report.[1] Measurements were first converted to weight percents on a dried solids basis and then converted to activities using the specific activity of each isotope.[11] The isotopes obtained from direct ICP-MS measurements included: Zr-93, Tc-99, Th-232, U-233, U-234, U-235, U-236, Np-237, U-238, Pu-239, Pu-240, and Pu-242.

2.1.2 Gamma Counting

Gamma Pulse Height Analysis (PHA) was performed on separate sub-samples of the peroxide fusion digestions of the Tank 40 – SB10 dried solids described in a previous report.[1] Detectors were carefully calibrated with known standards. Since detection efficiencies for gamma-rays vary with energy, they were determined for these specific radionuclide energies during the calibration process. The counting geometry was established during calibration and carefully duplicated for these measurements. Samples were diluted as necessary to achieve accurate counting. The isotopes obtained from Gamma PHA counting included Cs-134 (detection limit) and Cs-137.

2.2 Separation Methods

These analytical methods involved separation techniques that enabled radionuclides that were at low concentrations to be measured more accurately and to determine more reliable and lower detection limits. The techniques and methodology for these separations are maintained by the Nuclear Measurements (NM) group and will only be summarized here. Aliquots of the peroxide fusions or the aqua regia dissolutions were analyzed along with blanks. In all cases, the activity in the blanks did not contribute any significant activity to the radionuclides being analyzed. For the special cases involving Se-79, I-129 and Am/Cm aliquots of the sludge slurry were initially treated in the shielded cells for various separation steps (via the methodology detailed below) and then submitted to NM for further separation and/or counting.

2.2.1 Cl-36 Method

Aliquots of SB10 aqua regia dissolution were initially rendered caustic and subjected to two Monosodium Titanate (MST) and Crystalline Silico-Titanate (CST)-based decontamination steps. The resins and insoluble elements (i.e. actinides, lanthanides, strontium and yttrium) were filtered off, decontaminating the solution. The solutions were then acidified with nitric acid and further decontaminated with Bio-Rad AMP-1 (ammonium molybdophosphate) and Eichrom Diphonix resins. The Cl in the samples was subsequently precipitated as AgCl. The AgCl precipitate was counted using gas flow proportional counter analysis. The AgCl precipitate was then activated by neutron activation analysis to determine Cl losses during the processes. The HCl used to digest the samples initially was used to trace Cl-36 throughout the processes. The chlorine yields were used to correct Cl-36 results for any losses.

2.2.2 Ni-59/Ni-63 Method

This separation is based on isolation of Ni from the dissolved sludge using a column containing dimethylglyoxime (DMG) as an extractant that is specific for Ni. Each of the solutions resulting from the aqua regia digestions of the four samples of dried sludge slurry was spiked with a stable Ni carrier to trace the Ni separation and was then passed through a column containing the above extractant. The absorbed Ni was then eluted from each column. The Ni-59 was measured in the eluted solutions by its characteristic X-rays and Ni-63 by its beta particles. Total Ni in each eluted solution was measured by inductively coupled plasma atomic emission spectroscopy (ICP-AES). The radiochemical Ni analyses were corrected for the Ni carrier recoveries as measured by ICP-AES.

2.2.3 Gamma Counting Following Cs Removal

This method was used to determine Co-60, Ru-106, Ag-110m, Sb-125, Sn-126, Ba-133, Ce-144, Eu-152, Eu-154, Eu-155, and Am-241. These gamma emitters could not be determined directly because of the high Cs-137 activity in the samples. Consequently, the Cs-137 was removed. Aliquots of each of the four

peroxide fusions of the dried sludge slurry samples were treated with two batch additions of an ammonium phosphomolybdate resin to selectively remove the Cs-137 from the aliquots. This allowed gamma photons for isotopes at low concentrations to be detected or allowed lower detection limits to be determined for those isotopes that were not detected. A high purity coaxial germanium detector was utilized to detect the gamma rays from Co-60, Ru-106, Ag-110m, Sb-125, Sn-126, Ba-133, Ce-144, Eu-152, Eu-154, Eu-155, and Am-241. Only Co-60, Eu-154, and Am-241 were detected. Because of their low concentrations, the other isotopes were not detected. To obtain reliable and lower detection limits for these radionuclides, each of the solutions was counted for four hours or more. The detection limits were used to calculate the maximum activity of each for input to the projection calculations. Of this group of radionuclides, only Sn-126, Ba-133, and Am-241 have half-lives greater than 10 years. Even though the others have half-lives less than 10 years, their activities were included to calculate the total Curies present in SB10 at the selected decay times.

2.2.4 Se-79 Method

Four aliquots of wet sludge slurry were spiked with a known amount of stable Se as a carrier. The samples were digested with concentrated nitric acid. The Fe in the dissolutions was reduced to Fe(II) using ascorbic acid to ensure it would not interfere with subsequent decontamination steps designed to extract Y-90, the lanthanides and the actinides from the Se-traced dissolutions. The dissolutions were then treated with resins (Bio-Rad AMP-1, Eichrom Sr, RE, and Actinide resins) to reduce levels of Sr-90, Cs-137, Y-90, the lanthanides and the actinides to levels low enough to allow for their removal from the Shielded Cells and submission to NM. The Se traced decontaminated dissolutions were then further decontaminated with Bio-Rad AMP-1, Eichrom Sr and RE resin treatments. The total Se was reduced to Se metal using titanium (III) chloride, hydroxylamine hydrochloride, and ascorbic acid. The precipitated Se metal was then washed repeatedly with deionized water and dilute nitric acid. The Se metal was then dissolved with concentrated HBr, and the resulting $SeBr_4$ was extracted by solvent-solvent extraction using a tri-butyl phosphate/n-paraffin solvent extraction system. The Se was back extracted from the solvent. Aliquots of the purified Se fraction were then analyzed. A portion was neutron-activated in a Cf-252 neutron source at SRNL to determine the total amount of Se present in order to calculate the recovery of Se from the radiochemical separation. A second portion was counted by liquid scintillation to determine the Se-79 beta activity. The yields of the stable Se carrier were applied to the Se-79 beta activity result to determine Se-79 activities in the sample aliquots initially treated.

2.2.5 Sr-90 Method

Aliquots of each sample from the peroxide fusions were spiked with a stable Sr carrier and a stable Ce carrier. The Sr carrier was used for separation yielding purposes and the Ce carrier was used to enhance the separation rates of undesirable isotopes such as Y-90, the lanthanides, or the actinides. The spiked sample aliquots were initially oxidized using nitric acid. The Sr in the samples was extracted using commercially-available Sr extraction resin. This resin also extracts some of the Pu under the conditions used to extract the Sr. The Pu was washed from the resin using an oxalic acid/nitric acid mixture. The Sr was eluted from the resin, and the resulting solution concentrated. A portion of the purified Sr solution was neutron activated in a Cf-252 neutron activation facility at SRNL to determine the total Sr and to calculate the fraction of Sr isolated by the procedure. A second portion of each of the Sr fractions was stored for five to seven days to allow the grow-in of Y-90. Each fraction was then counted by liquid scintillation analysis to determine the Y-90 activity. The Sr-90 beta activity in each case was calculated from the Y-90 activity. The yields of the stable Sr carriers were applied to the Sr-90 beta activity results to determine Sr-90 activities in the original aliquots of the solutions resulting from the dissolution of the dried sludge slurry samples.

2.2.6 Pd-107 Method

Aliquots of each sample from the peroxide fusions were spiked with elemental Pd. Pd was then extracted from the samples using a DMG-based extractant. Pd-107 levels were measured using the ICP-MS, and the results were yielded from sample stable Pd recoveries as measured by the ICP-MS.

2.2.7 Sn-121m method

Aliquots of peroxide fusions were spiked with Sn-113. The aliquots were converted to a chloride matrix with hydrochloric acid, and elemental tin was extracted from the matrix using anion exchange. Tin was eluted with dilute nitric acid, and the tin extract was analyzed by low energy gamma-ray spectrometry for Sn-121m and for the Sn-113 tracer. The Sn-121m results were then yielded with the results from the Sn-113 recovery.

2.2.8 I-129 Method

The radionuclide I-129 is a long-lived beta-emitting fission product ($t_{1/2}$ = 1.6E+07 years) that is in SRS wastes. Aliquots of wet sludge slurry were spiked with a known amount of stable KI to act as an iodine tracer/carrier. The samples were digested with 8M nitric acid. The traced samples were then rendered caustic, precipitating out the actinides, lanthanides, Sr-90 and Y-90, among some other radioactive species. MST was added to further decontaminate the caustic dissolutions from Sr-90, Y-90, and select actinides and lanthanides. CST was added to reduce levels of Cs-137. The treated solutions were filtered, and the decontaminated filtrate was then removed from the Shielded Cells for submission to NM. The samples were decontaminated a final time with a resin treatment to remove Cs-137 and the actinide elements. The solution was then treated with $AgNO_3$ to precipitate the iodide ion as AgI. The precipitate was analyzed by low energy photon spectrometry to determine the amount of I-129 present. I-129 is detected by its characteristic gamma and x-ray emissions. The precipitate was then neutron-activated in a Cf-252 neutron source at SRNL to determine the total amount of iodine present to calculate the recovery of I-129 in the radiochemical separation.

2.2.9 Pm-147/Sm-151 Method

Aliquots of each sample from the peroxide fusions were spiked with a stable Sm carrier. The Sm carrier was used for separation yielding purposes. The spiked sample aliquots were initially oxidized using nitric acid. The Sm and Pm along with other trivalent species in the samples were extracted using Eichrom RE resin. The Sm and Pm where then extracted from the other radionuclides present using Eichrom Ln resin. A portion of the purified Pm/Sm solution was neutron activated in a Cf-252 neutron activation facility at SRNL to determine the total Sm and to calculate the fraction of Sm isolated by the procedure. A second portion of each of the Pm/Sm fractions was then counted by liquid scintillation analysis to determine the Pm-147 and Sm-151 activity. The Pm-147 measurement was conducted using a higher-energy beta window which was free of any interference from the low-energy Sm-151 beta. The Sm-151 beta result is corrected for any Pm-147 events occurring in its beta counting window when necessary. The yields of the stable Sm carriers were applied to the Sm-151 and the Pm-147 beta activity results to determine Sm-151 and Pm-147 activities in the original aliquots of the solutions resulting from the dissolution of the dried sludge slurry samples. A Pm-147 spiked sample was analyzed through the process to monitor and correct for any slight differences in the chemical recoveries of Sm and Pm.

2.2.10 Ra-226 Method

Aliquots of each sample from the peroxide fusions were decontaminated with a number of resin treatments. Radium was extracted from the matrix using an Empore-Radium-Disk-based extraction. Sample aliquots

of matrix spiked with Ra-226 were also run through the procedure for chemical yielding purposes. The disks containing extracted radium were then analyzed at three days following the extraction and at 14 days following the extraction to allow for daughter ingrowth. Sample Ra-226 results were corrected with matrix spike recoveries.

2.2.11 Th-229/-230, Ac-227 Method

Aliquots of each sample from the peroxide fusions were processed through a thorium separation procedure. The Th species were extracted from the matrix using two stages of a quaternary amine based solid phase extraction and purified further via co-precipitation with cerium. Th-229 and Th-230 concentrations were measured using passivated, implanted, planar silicon (PIPS) alpha spectrometers. The Th separation was yielded by measuring the Th-232 activities in the separated fractions, and comparing those activities to the Th-232 concentrations measured directly off aliquots of dissolution by ICP-MS. The Th-232 yields were used to correct the various analytes for any Th losses from the radiochemical separations.

2.2.12 Pa-231 Method

Aliquots of each replicate sample from the peroxide fusions were analyzed through a Pa separation procedure in duplicate with the duplicate containing a Pa-233 tracer. The dissolutions were decontaminated with AMP and quaternary amine-based resins. Protactinium species were then extracted from the matrix using a octylphenyl-N,N-di-isobutyl carbamoylphosphine oxide (CMPO)/tri-n-butylphosphate (TBP) based extractant. Pa-233 tracer concentrations were measured using high purity germanium spectrometers to determine separation yields. Pa-231 was measured using the ICP-MS. The Pa-233 tracer yields were decay-corrected and then used to correct the Pa-231 analyses for any losses from the radiochemical separations.

2.2.13 U-233, U-234, U-235, U-236 Method

To lower detection limits for U-233, U-234, U-235, and U-236 from those attainable from the direct ICP-MS analyses on the dissolution, aliquots of dissolution were purified with a diamyl, amylphosphonate (DAAP) based solid phase extraction. The purified aliquots were analyzed by the ICP-MS to measure U-233/U-238, U-234/U-238, U-235/U-238 and U-236/U-238 mass ratios. Those ratios were applied to U-238 concentrations measured by the ICP-MS directly on aliquots of dissolution to quantify the U-233, U-234, U-235 and U-236 concentrations.

2.2.14 Pu-238/-241 Method

Pu-241 is a beta-emitting Pu isotope that cannot be measured directly in the dissolved dried sludge slurry solutions because of its low concentration. Pu-241 has a relatively short half-life ($t_{1/2}$ = 15 years). Its concentration, along with that for Pu-238, was determined via peroxide fusion by isolating the Pu from each solution by a 2-thenoyltrifluoroacetone extraction procedure. The extracted Pu was then analyzed by beta and alpha counting to determine the ratio of beta activity from Pu-241 to the alpha activity from the other isotopes of Pu (Pu-238, Pu-239, Pu-240, and Pu-242). In the original dissolution solutions, the total alpha activity from the Pu isotopes was determined by alpha counting and ICP-MS. Knowing the total alpha activity from Pu in the solutions resulting from the extraction allows the concentration of Pu-241 in the original dissolution solutions to be calculated using the beta/alpha ratio determined in the extracted solution. In the extracted solution, the alpha counting technique also gives the alpha counts due specifically to Pu-238 so that the total amount of Pu-238 can be determined. The activities of these two radionuclides were then used in the calculations to determine the reportable radionuclides.

2.2.15 Pu-242/Pu-244 Method

The plutonium from aliquots of sludge batch slurry peroxide fusion dissolutions were extracted using Eichrom TEVA resin. The Pu-containing extracts were then analyzed by ICP-MS to determine Pu-239, Pu-240, Pu-242, and Pu-244 masses. The masses were converted to activities and ratioed to the sum of the Pu-239 and Pu-240 activities from the ICP-MS measurement. Those ratios were then multiplied by the measured Pu-239/240 result of the 2-thenoyltrifluoroacetone (TTA) analysis to quantify sample Pu-239, Pu-240, Pu-242, and Pu-244 activities.

2.2.16 Am/Cm Method

This method was used for Am-241, Am-242m, Cm-242, Am-243, Cm-243, Cm-244, Cm-245, Cm-246, Cm-247, Cm-248, Bk-247, Cf-249, Cf-250, Cf-251, and Cf-252. These radionuclides are neutron activation products produced in the SRS reactors. These isotopes are difficult to measure because of their low concentrations in the sludge slurry and the dilutions necessary to get the dissolved slurry samples out of the Shielded Cells. Of these isotopes, the Am-241 can be easily and accurately analyzed directly by extended gamma counting of the dissolved sludge (see Section 2.2.3). For the other radionuclides listed above, a separation method has been developed by NM for isolating Am, Cm, Bk, and Cf from a wet sludge slurry solution. The slurry is digested in the Shielded Cells with a peroxide fusion. The trivalent actinides are then extracted from the dissolution filtrate using a commercially-available ion exchange resin (Eichrom DGA). As Y-90 coextracts with the trivalent actinides on DGA resin, the treated samples were held in the Shielded Cells for nine days to allow the Y-90 to decay before they were removed and submitted to NM. The solutions were purified further with a second RE resin extraction followed by an Eichrom Ln resin extraction. The Am, Cm, Bk, and Cf extracts were then analyzed by alpha and low-energy gamma counting techniques as well as by ICP-MS. The radionuclides Cm-242, Am-242m, Cm-244, and Cf-252 were measured by alpha spectroscopy, Am- 241, Am-243, Cm-243, Cm-245, Cf-249, and Cf-251 were measured by low-energy gamma spectroscopy, and Cm-245, Cm-246, Cm-247, Bk-247, Cm-248, and Cf-250 were measured by ICP-MS. The fraction of each actinide element isolated by this ion exchange technique was determined by comparing the measured concentrations of Am-241 in the eluted solutions with their respective concentration in the original dissolved slurry that was measured by direct gamma counting of Cs-137 removed aliquots of the dissolved slurry.

By using this technique, the radionuclides Am-242m, Am-243, Cm-242, Cm-244, Cm-245, and Cm-246 were detected and measured along with the Am-241. All the other radionuclides (Cm-243, Cm-247, Bk-247, Cm-248, Cf-249, Cf-250, Cf-251 and Cf-252) had concentrations below the detection limit of the analytical methods. For the later series of radionuclides, the detection limits were then used as the maximum concentrations or activities that could be present.

2.3 Calculated Radionuclide Activities

2.3.1 Nb-93m

The radionuclide Nb-93m ($t_{1/2}$ = 16.1 years) is in SRS HLW as the decay product of the radioactive fission product Zr-93 ($t_{1/2}$ = 1.53E+06 years). For previous sludge batches both the Zr-93 and Nb-93m became reportable after the waste was ~100 years old.[12,13,14,15,16,17,8,18,9,10] The concentration of Nb-93m during SB10 vitrification can be calculated if the age of the sludge and the concentration of Zr-93 are known. The age of the waste can be calculated from the measured concentration at mass 90 that is composed of Sr-90 ($t_{1/2}$ = 28.5 years) and its daughter Y-90 ($t_{1/2}$ = 2.67 days) that decays to stable Zr-90. The radionuclide Sr-90 is the initial long lived radioactive product of the isobaric decay chain at mass 90. In the ICP-MS analysis of the aqua regia dissolutions of the SB10 total solids, the concentration at mass 90 which is composed of Sr-90, Y-90, and Zr-90 was 0.0180 wt%. The concentration of Sr-90 determined by radioactive counting was

0.00430 wt% (see Section 2.2.5). The ratio of these two values representing the fraction of Sr-90 that remained in this sludge is 0.239. With the half-life of Sr-90 and standard equation for radioactive decay, the age of the sludge can be calculated to be 59 years.

The Zr-93 activity in Tank 40 SB10 WAPS measured by ICP-MS is 0.595 µCi/g. As Zr-93 decays, approximately 97% of transformations result in the production of Nb-93m. Because of the large difference in the half-lives of Zr-93 and its radioactive daughter Nb-93m, these radionuclides are in secular equilibrium. Consequently, the Nb-93m activity can be calculated from the age of the waste using the following relationship:

$$A_{Nb-93m}(t) = 0.97 * A_{Zr-93,t=0} * e^{-\frac{0.693t}{1.53E6}} * \left[1 - e^{-0.693t/16.1}\right] \quad \text{(Eq. 1)}$$

Where $A_{Zr\text{-}93,\,t=0}$ is the Zr-93 activity at time zero (equivalent to current activity, due to long half-life of Zr-93), and t is the age of the waste in years.

2.3.2 Cd-113m

With a half-life of 13.7 years and specific activity of 217 Ci/g,[11] Cd-113m may also qualify as a WAPS-reportable radionuclide. However, Cd-113m primarily decays by ß emission and thus would require a careful separation technique to measure in sludge slurry. Also, the determination of Cd-113m by ICP-MS of the dissolved dried solids is essentially impossible because of the presence of natural Cd in the sludge. Natural Cd is 12.3 % Cd-113 with a half-life greater than 1.1E+11 years. This makes its activity negligible at all times. Finally, the fission yield of Cd-113m is very small, 1.66E-04%, and thus its concentration is expected to be very small in the HLW sludge.

An upper limit of the mass concentration of Cd-113m can be estimated by using the fission yield scaling factor (FYSF).[19] The FYSF relates the concentration of a fission product in the total sludge solids to the fission yield and the atomic mass of that fission product. The atomic mass of that isotope has to be included in the equation because fission yields are given in terms of atoms produced per 100 fissions of U-235 and not in terms of mass percent of the isotope produced. The equation for the concentration in weight percent is:

$$\text{Concentration (wt\%)} = \text{FYSF (fission yield x atomic mass)} \quad \text{(Eq. 2)}$$

Thus, the FYSF for each measured isotope can be calculated from the following:

$$FYSF_i = wt\%_i \,/\, (FY_i \times AM_i) \quad \text{(Eq. 3)}$$

Where $FYSF_i$ = the fission yield scaling factor based on isotope i,
$wt\%_i$ = the weight per percent of isotope i in the HLW total dried solids,
FY_i = the fission yield of isotope i, and
AM_i = the atomic mass of isotope i.

Several of the U-235 fission products have the six critical chemical and nuclear properties that allow calculation of a constant FYSF for a particular sludge. These properties have been previously discussed.[19] In SB10, there are 12 isotopes that have these six properties. The FYSFs calculated for these 12 isotopes are presented in the Table 2-1. The measured concentrations in the Table 2-1 were determined by ICP-MS analysis of four samples of dissolved SB10 total dried solids by the aqua regia method. The average of the FYSFs calculated for the 12 fission products is 3.95E-05 with a relative standard deviation of 14%.

Table 2-1. SB10 Calculated FYSF Values

FYSF	wt% Total Solids	Fission Yield	FYSF
Ru-101	2.07E-02	5.20	3.94E-05
Ru-102	1.82E-02	4.30	4.16E-05
Rh-103	9.70E-03	3.03	3.11E-05
Ru-104	9.52E-03	1.88	4.87E-05
La-139	3.85E-02	6.41	4.33E-05
Pr-141	3.29E-02	5.80	4.02E-05
Nd-143	2.94E-02	5.96	3.45E-05
Nd-144	3.45E-02	5.50	4.36E-05
Nd-145	2.03E-02	3.93	3.55E-05
Nd-146	1.77E-02	3.00	4.04E-05
Sm-147	1.04E-02	2.25	3.13E-05
Sm-148	1.09E-02	1.67	4.40E-05
		Average	3.95E-05

With the FYSF for SB10, the maximum possible concentration in terms of weight percent of dried solids can be estimated for the other U-235 fission products. These estimations are a maximum for the other fission products because they do not have the necessary six properties as the fission products given in Table 2-1. For example, the radionuclide Cd-113m has a very large neutron absorption cross section (20,000 barns), and thus it was transmuted in the reactors at SRS to stable Cd-114 while the reactors were in operation. Consequently, the concentration calculated for Cd-113m with Equation 2 and a fission yield of 1.66E-04% can be considered the maximum concentration and is estimated below:

$$\text{Concentration Cd-113m (wt\%)} = 3.95\text{E-}05 \times 1.66\text{E-}04 \times 113 = 7.40\text{E-}07 \text{ wt\%}.$$

2.3.3 Cs-135

By using the ratio of Cs-135 to Cs-137 in the supernate, and the amount of Cs-137 in the sludge slurry, the activity for Cs-135 in the sludge slurry can be calculated. As opposed to the solids analysis, the supernate analysis was used because of Ba-135 interferes with the detection of Cs-135 in the solid. Ba-135 is a stable isotope that is insoluble; therefore, no Ba-135 would interfere with the Cs-135 signal from the supernate. This calculated value for Cs-135 in SB10 was performed using the measured ICP-MS values for mass 135 and 137 from the supernate equal to 8.69E+01 µg/L and 1.84E+02 µg/L, respectively, giving a mass 135: mass 137 ratio of 0.4717. This ratio was used to calculate the Cs-135 in the sludge slurry by multiplying by the measured Cs-137 in the sludge slurry, giving 0.4717 x 4.63E-04 wt% = 2.18E-04 wt% Cs-135.

2.3.4 Pb-210 and Ra-226

Calculation of the quantities of radioactive decay products Pb-210 and Ra-226 were performed based on the calculated value that the waste is 59 years old. Two of the parent radionuclides (U-234 and Pu-238) used in these calculations are indicated below in Table 3-1 of Section 3.1. The third parent radionuclide, Th-230 is long-lived alpha-emitting radionuclide ($t_{1/2}$ = 7.5E+04 years) that is a decay product of U-234 and Pu-238.

In calculating the concentrations of decay products, the first step was to calculate the concentrations of the parent radionuclides 59 years ago. Utilizing the standard activity decay relationship, the parent concentration in 1964 was calculated as the product of the current concentration (in 2023) and the term [exp(0.693*59/ $t_{1/2}$)], where $t_{1/2}$ is the half-life of the parent radionuclide in units of years. The second step was to determine the decay product concentration in 2023, utilizing a decay/ingrowth calculation to quantify the progeny arising from 59 years of decay of the 1964 parent concentrations. RadDecay® v7.00 software (see Section 3.3 below) was utilized for these calculations.[20]

The calculated Ra-226 activity was three orders of magnitude lower than the detection limit measured for this radionuclide; therefore, the calculated value was employed in the decay calculations so as not to artificially inflate the contribution from this radionuclide.

2.4 Assumptions

The following assumptions were used in determining the radionuclide inventory inputs for the RadDecay® v 7.00 calculations:

1. When a radionuclide analysis yielded results that were below the detection limits for the measurement technique, the detection limit was used as the initial activity in the RadDecay® v 7.00 input.
2. Yttrium-90 was assumed to be in secular equilibrium with Sr-90.
3. Californium-250 and Californium-252 were assumed to have the same activity as Cm-242. The primary alpha radiation from these radionuclides have nearly identical emission energies and could not be resolved by the detection method. The measured Cm-242 activity was used as the initial activity for Cm-242, Cf-250, and Cf-252.
4. The amount of Ba-137m was calculated from the decay of Cs-137 wherein the Cs-137 radionuclide decays to the metastable Ba-137m approximately 94% of the time. Thus, the amount of Ba-137m was assumed to be 0.946*A_o^{Cs-137} where A_o^{Cs-137} is the initial amount of Cs-137.
5. The amounts of Cs-135, Nb-93m, Cd-113m, Pb-210, and Ra-226 were calculated based on methodology given in Section 2.3.

The assumptions were made to provide a conservative estimate of the radionuclide inventory for the Tank 40 SB10 WAPS sample.

2.5 Quality Assurance

This work was performed under a TTR.[6] The analysis herein satisfies the radionuclide composition portion of Task 3, Item 2 of the TTQAP associated with this TTR.[7] The TTR identifies these results as waste form affecting and complies with the requirements of RW-0333P. The TTR identifies the Functional Classification as Safety Class. Thus, this document was reviewed by Design Verification by Document Review. The requirements for performing reviews of technical reports and the extent of review are established in manual E7 2.60. SRNL documents the extent and type of review using the SRNL Technical Report Design Checklist contained in WSRC-IM-2002-00011, Rev. 2. Data are recorded in the electronic laboratory notebook K7482-00430-28. Calculations performed using the RadDecay® v. 7.00 software are performed in accordance with software quality assurance plan.[21]

3.0 Results and Discussion

3.1 Summary of Radionuclide Activities for Input

The radionuclide composition of the SB10 Tank 40 WAPS sample (HTF-40-23-24) is reported for waste acceptance purposes in Table 3-1. The radionuclide activities in Table 3-1 were used as inputs for decay calculations. Table 3-1 gives for each radionuclide the associated specific activity in units of Ci/g,[11,10, 22] concentration in wt% of the total solids, activity in µCi/g of total dried solids, and activity in Ci/gal of slurry, and the associated relative standard deviation of the replicate analyses. The % relative standard deviations (%RSD) are given for the quadruplicate analyses unless otherwise indicated in the footnotes of Table 3-1. Activity measurements that were below the detection limit of a given technique or calculated do not have an associated relative standard deviation and are marked as 'NA' in Table 3-1. The analytical method for each of the radionuclides is also shown. The assay date for the radionuclides in Table 3-1 was March 20, 2023, which is also the start date for the index calculations.

The total measured alpha activity of the digested samples was 1.19E+02 µCi/g of total dried solids and the total measured beta activity was 1.41E+04 µCi/g of total dried solids.

Table 3-1. SB10 Radionuclides and Activities Used as Inputs in RadDecay® v. 7.00

Nuclide	Specific Activity (Ci/g)	wt% Total Solids	Activity (µCi/g dried solids)*	Ci/gal-Slurry	%RSD	Analytical Method
Cl-36	3.30E-02	5.25E-05	1.73E-02	1.09E-05	24	Cl-36
Ni-59	8.08E-02	3.20E-04	2.59E-01	1.63E-04	42	Ni-59/-63
Co-60	1.13E+03	8.43E-09	9.54E-02	6.03E-05	5.5	Cs-removed gamma counting
Ni-63	6.17E+01	5.23E-05	3.22E+01	2.04E-02	15	Ni-59/-63
Se-79	6.97E-02	1.03E-06	7.17E-04	4.53E-07	40	Se-79
Sr-90	1.36E+02	4.30E-03	5.87E+03	3.71E+00	4.1	Sr-90
Y-90	5.44E+05	1.08E-06	5.87E+03	3.71E+00	4.1	Secular Equilibrium w/ Sr-90
Zr-93	2.51E-03	2.37E-02	5.95E-01	3.76E-04	23	ICP-MS
Nb-93m	2.83E+02	1.88E-07	5.31E-01	3.36E-04	NA	obtained from Zr-93 & est. waste age
Tc-99	1.70E-02	1.51E-03	2.57E-01	1.62E-04	2.1	ICP-MS
Ru-106	3.35E+03	< 3.66E-09	< 1.22E-01	< 7.73E-05	NA	Cs-removed gamma counting
Pd-107	5.14E-04	< 9.11E-05	< 4.68E-04	< 2.96E-07	NA	Pd-107
Cd-113m	2.17E+02	7.40E-07	1.61E+00	1.01E-03	NA	estimated based on fission yield
Sn-121m	5.91E+01	3.47E-07	2.05E-01	1.30E-04	8.7	Sn-126/-121m
Sb-125	1.03E+03	< 9.13E-09	< 9.43E-02	< 5.96E-05	NA	Cs-removed gamma counting
Te-125m	1.80E+04	< 5.23E-10	< 9.43E-02	< 5.96E-05	NA	Secular Equilibrium w/ Sb-125
Sn-126	2.84E-02	< 9.67E-06	< 2.74E-03	< 1.74E-06	NA	Sn-126/-121m
I-129	1.77E-04	1.86E-03	3.29E-03	2.08E-06	26	I-129
Ba-133	2.50E+02	< 2.04E-08	< 5.10E-02	< 3.22E-05	NA	Cs-removed gamma counting
Cs-134	1.29E+03	< 8.55E-08	< 1.11E+00	< 7.00E-04	NA	gamma
Cs-135	1.15E-03	2.18E-04	2.51E-03	1.59E-06	NA	ratio of Cs-135/Cs-137 supernate applied
Cs-137	8.70E+01	4.63E-04	4.03E+02	2.55E-01	1.4	gamma
Ba-137m	5.38E+08	7.08E-11	3.81E+02	2.41E-01	1.4	0.946 x Cs-137 activity
Ce-144	3.19E+03	< 1.62E-08	< 5.16E-01	< 3.26E-04	NA	Cs-removed gamma counting
Pm-147	9.27E+02	< 4.28E-06	< 3.97E+01	< 2.51E-02	NA	Pm-147/Sm-151
Sm-151	2.63E+01	2.50E-04	6.57E+01	4.15E-02	21	Pm-147/Sm-151
Eu-154	2.70E+02	1.45E-06	3.90E+00	2.47E-03	13	Cs-removed gamma counting
Eu-155	4.65E+02	< 4.26E-08	< 1.98E-01	< 1.25E-04	NA	Cs-removed gamma counting
Pb-210	7.63E+01	4.00E-12	3.06E-06	1.93E-09	NA	calculated from U series and est. waste age
Ra-226	9.89E-01	< 1.03E-07	< 1.02E-03	< 6.45E-07	NA	Ra-226
Ra-226 (calc)	9.89E-01	5.81E-10	5.74E-06	3.63E-09	NA	calculated from U series and est. waste age
Ac-227†	7.23E+01	1.88E-09	1.36E-03	8.61E-07	44	Th-229/-230
Th-229	2.13E-01	1.31E-07	2.78E-04	1.76E-07	47	Th-229/-230
Th-230‡	2.11E-02	1.18E-06	2.50E-04	1.58E-07	NA	Th-229/-230
Pa-231	4.72E-02	< 9.03E-07	< 4.27E-04	< 2.70E-07	NA	Pa-231
Th-232	1.10E-07	1.42E+00	1.56E-03	9.83E-07	2.8	ICP-MS
U-233	9.68E-03	4.98E-04	4.83E-02	3.05E-05	4.4	U-233/-234/-235/-236
U-234	6.25E-03	1.41E-03	8.80E-02	5.56E-05	3.4	U-233/-234/-235/-236
U-235	2.16E-06	3.19E-02	6.89E-04	4.36E-07	5.2	U-233/-234/-235/-236
U-236	6.47E-05	2.14E-03	1.39E-03	8.76E-07	5.2	U-233/-234/-235/-236
Np-237	7.05E-04	1.69E-03	1.19E-02	7.54E-06	3.3	ICP-MS
U-238	3.36E-07	2.48E+00	8.33E-03	5.26E-06	4.2	ICP-MS
Pu-238	1.71E+01	5.89E-04	1.01E+02	6.37E-02	8.2	Pu-238/-241
Pu-239	6.22E-02	1.64E-02	1.02E+01	6.45E-03	8.4	Pu-242/-244
Pu-240	2.28E-01	1.35E-03	3.08E+00	1.94E-03	8.7	Pu-242/-244
Pu-241	1.03E+02	3.31E-05	3.41E+01	2.15E-02	11	Pu-238/-241
Am-241	3.43E+00	3.62E-04	1.24E+01	7.85E-03	7.5	Cs-removed gamma counting
Pu-242	3.82E-03	1.06E-04	4.05E-03	2.56E-06	8.7	Pu-242/-244
Am-242m§	9.72E+00	1.27E-07	1.23E-02	7.79E-06	8.2	AmCm
Cm-242§	3.31E+03	3.08E-10	1.02E-02	6.44E-06	8.5	AmCm
Am-243	1.99E-01	3.87E-05	7.72E-02	4.88E-05	10	AmCm
Cm-243	5.16E+01	< 1.06E-07	< 5.49E-02	< 3.47E-05	NA	AmCm
Cm-244	8.09E+01	3.66E-06	2.96E+00	1.87E-03	8.9	AmCm
Cm-245	1.72E-01	4.29E-07	7.37E-04	4.66E-07	8.6	AmCm
Cm-246	3.07E-01	6.26E-07	1.92E-03	1.22E-06	14	AmCm
Cm-247	9.28E-05	< 1.28E-08	1.19E-08	< 7.52E-12	NA	AmCm
Bk-247	1.05E+00	< 1.27E-08	< 1.33E-04	< 8.39E-08	NA	AmCm
Cm-248	4.25E-03	< 1.13E-07	< 4.80E-06	< 3.04E-09	NA	AmCm
Cf-249	4.09E+00	< 5.35E-08	< 2.19E-03	< 1.38E-06	NA	AmCm
Cf-250	1.09E+02	<9.32E-09	< 1.02E-02	<6.44E-06	NA	AmCm
Cf-251	1.59E+00	< 3.73E-07	< 5.91E-03	< 3.73E-06	NA	AmCm
Cf-252	5.38E+02	<1.89E-09	< 1.02E-02	<6.44E-06	NA	AmCm

* Less than values represent the minimum detection limit value. † Ac-227 was calculated with three replicates. ‡ Th-230: only one of four replicates was above detection limit. § Am-242m and Cm-242 were calculated with two replicates.

3.2 Additional Requested Radionuclides

Additional radionuclides not included as inputs in radionuclide decay calculations are given in Table 3-2.

Table 3-2. List of Additional Radionuclides and Activities

Nuclide	Specific Activity (Ci/g)	wt% Total Solids	Activity (µCi/g dried solids)*	Ci/gal-Slurry	%RSD	Analytical Method
Ag-110m	4.75E+03	< 5.74E-10	< 2.73E-02	< 1.72E-05	NA	Cs-removed gamma counting
Pr-144	7.56E+07	< 6.83E-13	< 5.16E-01	< 3.26E-04	NA	Secular Equilibrium w/ Ce-144
Pr-144m	1.81E+08	< 2.84E-13	< 5.16E-01	< 3.26E-04	NA	Secular Equilibrium w/ Ce-144
Eu-152	1.73E+02	< 2.78E-08	< 4.80E-02	< 3.04E-05	NA	Cs-removed gamma counting
Pu-244	1.77E-05	<3.72E-05	<6.59E-06	< 4.17E-09	NA	Pu-242-244
Rh-106	3.56E+09	<3.44E-15	<1.22E-01	< 7.73E-05	NA	Secular Equilibrium w/ Ru-106
H-3†	9.65E+03	NA	NA	< 2.76E-05	NA	Tritium

*Less than value represent the minimum detection limit. † Tritium was determined on an undigested slurry.

3.3 Determination of Reportable Radionuclides

The commercially-available software RadDecay® v. 7.00[20] was used to identify reportable radionuclides through calendar year 3115 at the specified index years. The initial activities were entered into RadDecay® based on the analytical results of the Tank 40 SB10 WAPS samples as given in Table 3-1. The assay date for the radionuclide analysis was March 20th, 2023. This day was used as the initial date in the decay calculations as directed by DWPF.[3] Some of the radionuclide analyses were completed after this date. These were not decay-corrected since the radionuclides either have a long half-life or are in secular equilibrium with a radionuclide with a long half-life. Calculations were performed in 100-year intervals except for the initial interval, which was 92 years per the aforementioned directive. The concentrations of all radionuclides present at the various index years are given in Appendix C. The ICRP-107 library was used for half-life data for the decay calculations.[23]

The RadDecay® v 7.00 software was benchmarked against Scale 6.1,[24] which is a modeling and simulation suite for nuclear safety analysis and design. The RadDecay® v. 7.00 software proved acceptable for WAPS compliance use.[25]

The total activity of the dried sludge as input into the RadDecay® calculations at year 2023 was 1.28E+04 µCi/g. The activity of only the radionuclides that were reportable at year 2023 was 6.54E+03 µCi/g. The total activity of the sludge in 3115 is 2.71E+01 µCi/g, of which 1.71E+01 µCi/g is coming from reportable radionuclides. The discrepancies between the total activity of the sludge batch and the activity attributable to reportable radionuclides is caused by radionuclides with half-lives shorter than 10 years that have a significant contribution to the total activity. The total activity and reportable radionuclide activity of SB10 is shown in Figure 3-1.

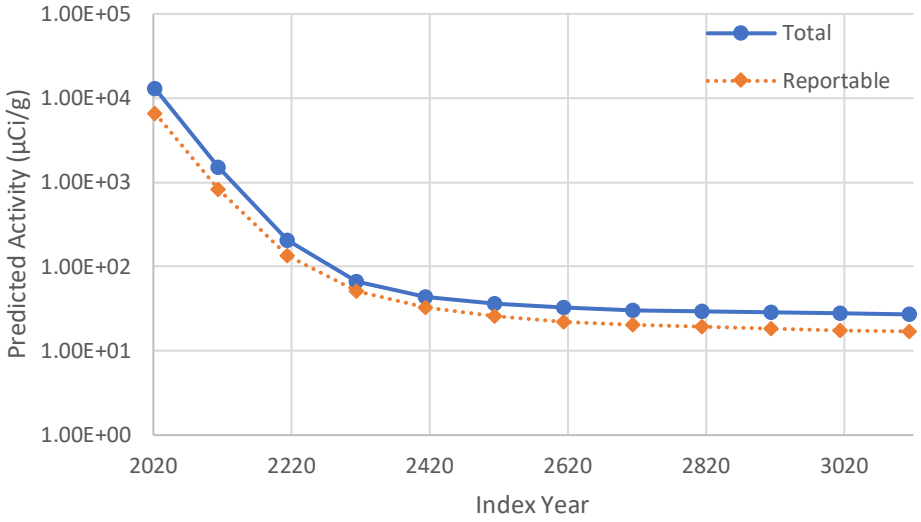


Figure 3-1. Total and Reportable Radionuclides Activity of DWPF SB10 Tank 40 Sample, as Measured for 2023 and Predicted for Index Years Through 3115 by RadDecay® v. 7.00. The Discrepancy Between the Total and Reportable Activity is Attributable to Radionuclides with Half-lives of Less than 10 Years.

As described in the WCP and WQR, DWPF complies with DOE WAPS requirements by considering the half-life requirement ($t_{1/2}$ > 10 years) and radionuclides with concentrations greater than 0.01% of the total inventory from the approximate time of production through 1100 years. Twenty-six radionuclides were identified in the Tank 40 sample using these criteria as reportable for SB10. The twenty-six reportable radionuclides are given in Table 3-3. The blue shaded cells in the Table 3-3 indicate in which index year the radionuclides are reportable. Appendix A gives plots of the reportable radionuclides through the index years as a function of their activities and their % of the total activity. SB10 reportable radionuclides are compared to previous sludge batches in Appendix B. All of the reportable radionuclides from the SB10 Tank 40 sample are radionuclides that were reportable in two or more previous sludge batches.

Table 3-3. DWPF SB10 Reportability Matrix for 2023 to 3115 – Blue Shaded Cell Indicates The Radionuclide Is Reportable For That Year.

Isotope	2023	2115	2215	2315	2415	2515	2615	2715	2815	2915	3015	3115
Cl-36												
Ni-59												
Ni-63												
Sr-90												
Zr-93												
Nb-93m												
Tc-99												
Cd-113m												
Sn-126												
I-129												
Cs-137												
Sm-151												
Th-229												
U-233												
U-234												
Np-237												
U-238												
Pu-238												
Pu-239												
Pu-240												
Pu-241												
Am-241												
Pu-242												
Am-243												
Cm-244												
Cf-251												

3.4 The Ratio by Weight of U and Pu Isotopes

The mass ratios for uranium and plutonium content must be reported for IAEA safeguards as specified in WAPS 1.6. As such, these uranium and plutonium radionuclides are presented in this section. Tables 3-4 and 3-5 show the distribution of uranium and plutonium isotopes determined for SB10 from the Tank 40 WAPS sample.

Table 3-4. DWPF SB10 Uranium Isotopes Distribution

Isotope	wt% Total Solids	Percent Distribution
U-233	4.98E-04	0.0198
U-234	1.41E-03	0.0561
U-235	3.19E-02	1.27
U-236	2.14E-03	0.0853
U-238	2.48E+00	98.6
Total	2.51E+00	100

Table 3-5. DWPF SB10 Plutonium Isotopes Distribution

Isotope	wt% Total Solids	Percent Distribution
Pu-238	5.89E-04	3.18
Pu-239	1.64E-02	88.8
Pu-240	1.35E-03	7.30
Pu-241	3.31E-05	0.179
Pu-242	1.06E-04	0.573
Total	1.85E-02	100

The uranium isotopes U-233, U-234, and U-238, as well as all Pu isotopes, are reportable since they meet the criteria in WAPS 1.2 of having half-lives greater than 10 years and a contribution to the overall activity of greater than 0.01%. U-235 and U-236 are also reportable to comply with WAPS 1.6.

4.0 Conclusions

The following twenty-six radionuclides were identified as reportable for the SB10 Tank 40 sample using the criteria set forth by WAPS 1.2 and WQR:

Cl-36	Ni-59	Ni-63	Sr-90	Zr-93
Nb-93m	Tc-99	Cd-113m	Sn-126	I-129
Cs-137	Sm-151	Th-229	U-233	U-234
Np-237	U-238	Pu-238	Pu-239	Pu-240
Pu-241	Am-241	Pu-242	Am-243	Cm-244
Cf-251				

Additionally, U-235 and U-236 are also reportable to comply with IAEA safeguards. Taking these gives a total of twenty-eight reportable radionuclides for SB10 Tank 40 sample. Additional radionuclides may be reportable from the salt streams added during the SB processing at DWPF.

5.0 References

1. Hunter, S. C. *Chemical Characterization Results for Tank 40 Sludge Batch 10 Waste Acceptance Product Specifications (WAPS) Sample*; SRNL-STI-2023-00302, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2024.

2. *Waste Acceptance Product Specifications for Vitrified High-Level Waste Forms*; DOE/EM-0093, Rev. 2; Department of Energy - Office of Environmental Management: 1996.

3. Brotherton, K. M.; Ray, J. W. *Defense Waste Processing Facility (DWPF) Path Forward on DOE-EM Waste Acceptance Product Specifications (WAPS) 2015 Index Year for Radionuclide Reporting*; SRR-WSE-2018-00020, Rev. 0; 2018.

4. *DWPF Waste Form Compliance Plan*; WSRC-IM-91-116-0, Rev. 14; Savannah River Mission Completion: Aiken, SC, 2023.

5. Ray, J. W.; Culbertson, B. H.; Edwards, T. B.; Bibler, N. E.; Marra, S. L. *Reporting the Radionuclide Inventory of the DWPF Product*; WSRC-IM-91-116-4, Rev. 4; 2018.

6. Russell, K. J. *Sludge Batch 10 Qualification, Confirmatory, and Waste Acceptance Product Specifications Samples*; X-TTR-S-00075, Rev. 3; Savannah River Remediation: Aiken, SC, 2021.

7. Pareizs, J. M.; Martino, C. J. *Task Technical and Quality Assurance Plan for Sludge Batch 10 Sample Analysis and Qualification*; SRNL-RP-2019-00658, Rev. 2; Savannah River National Laboratory: Aiken, SC, 2021.

8. Reboul, S. H.; DiPrete, D. P.; Click, D. R.; Bannochie, C. J. *Reportable Radionuclides in DWPF Sludge Batch 7a (Macrobatch 8)*; SRNL-STI-2011-00720, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2011.

9. Bannochie, C. J.; DiPrete, D. P. *Determination of Reportable Radionuclides for DWPF Sludge Batch 8 (Macrobatch 10)*; SRNL-STI-2014-00179, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2014.

10. Trivelpiece, C. L.; Kubilius, W. P.; Diprete, D. P. *Determination of Reportable Radionuclides for Defense Waste Processing Facility (DWPF) Sludge Batch 9 (Macrobatch 11)*; SRNL-STI-2018-00680, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2019.

11. *Integrated Data Base Report - 1996: U.S. Spent Nuclear Fuel and Radioactive Waste Inventories Projections and Characteristics*; DOE/RW-0006, Rev. 13; Oak Ridge National Laboratory: Oak Ridge, TN, 1997.

12. Fellinger, T. L.; Bibler, N. E.; Harbour, J. R. *Characterization of and Waste Acceptance Radionuclides to be Reported For DWPF Macro Batch 2 (ESP 215-ESP 221)*; WSRC-RP-99-00436, Rev. 1; Westinghouse Savannah River Company: Aiken, SC, 2004.

13. Bibler, N. E.; Diprete, D. P.; Harbour, J. R. *Determination of Reportable Radionuclides for DWPF Sludge Batch 2 (Macro Batch 3)*; WSRC-TR-2002-00255; Westinghouse Savannah River Company: Aiken, SC, 2002.

14. Bannochie, C. J.; Bibler, N. E. *Determination of Reportable Radionuclides For DWPF Sludge Batch 3 (Macrobatch 4)*; WSRC-TR-2005-00157, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2005.

15. Bannochie, C. J.; Bibler, N. E.; Diprete, D. P. *Determination of Reportable Radionuclides for DWPF Sludge Batch 4 (Macrobatch 5)*; WSRC-STI-2008-00142, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2008.

16. Bannochie, C. J.; Bibler, N. E.; Diprete, D. P. *Determination of Reportable Radionuclides For DWPF Sludge Batch 5 (Macrobatch 6)*; SRNL-STI-2009-00821, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2010.

17. Bannochie, C. J.; Diprete, D. P. *Determination of Reportable Radionuclides For DWPF Sludge Batch 6 (Macrobatch 7)*; SRNL-STI-2011-00189, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2011.

18. Crawford, C. L.; DiPrete, D. P. *Determination of Reportable Radionuclides for DWPF Sludge Batch 7b (Macrobatch 9)*; SRNL-STI-2012-00294, Rev. 2; Savannah River National Laboratory: Aiken, SC, 2014.

19. Bibler, N. E.; Fellinger, T. L.; Hobbs, D. T. *Technetium-99 Behavior in Savannah River Site High Level Waste Sludges During Waste Processing*; WSRC-MS-2004-00614, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2004.

20. *RadDecay*; Version 7.00; Grove Software: Lynchburg, VA 24503.

21. Hunter, S. C. *Software Quality Assurance Plan for Commercial Radioactive Decay Calculation Package RadDecay® v. 7.00 Utilized by Chemical Flowsheet Development*; G-SQP-A-00006, Rev. 7; Savannah River National Laboratory: Aiken, SC, 2024.

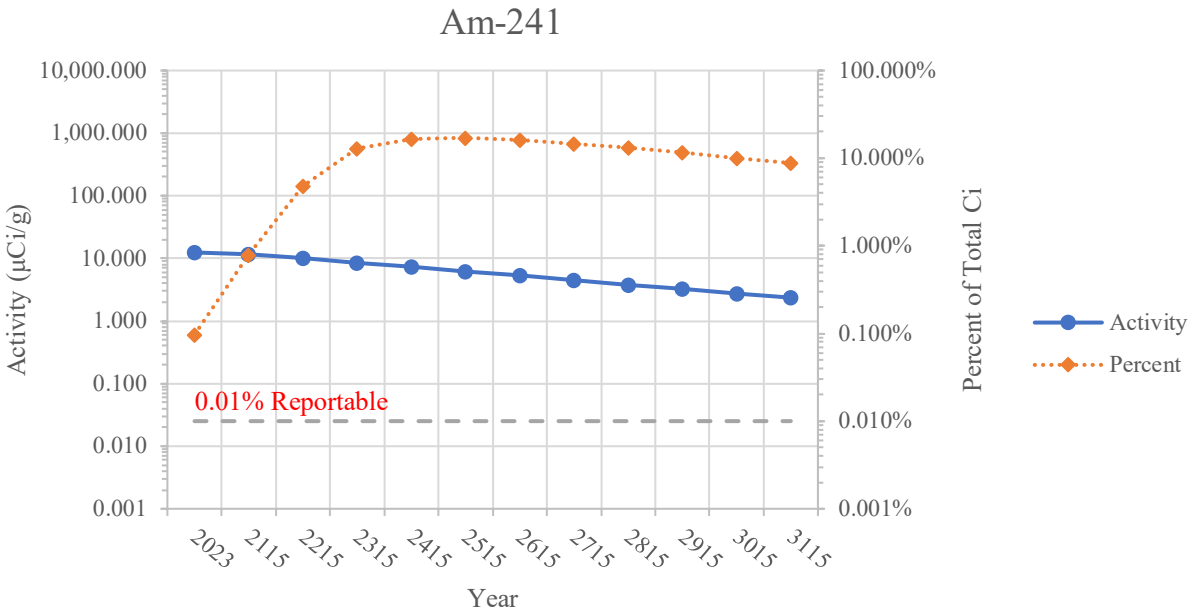
22. Boggs, B. *Compilation of Data on Radionuclide Data for Specific Activity, Specific Heat and Fission Product Yields*; WSRC-TR-99-00344; Westinghouse Savannah River Company: Aiken, SC, 1999.

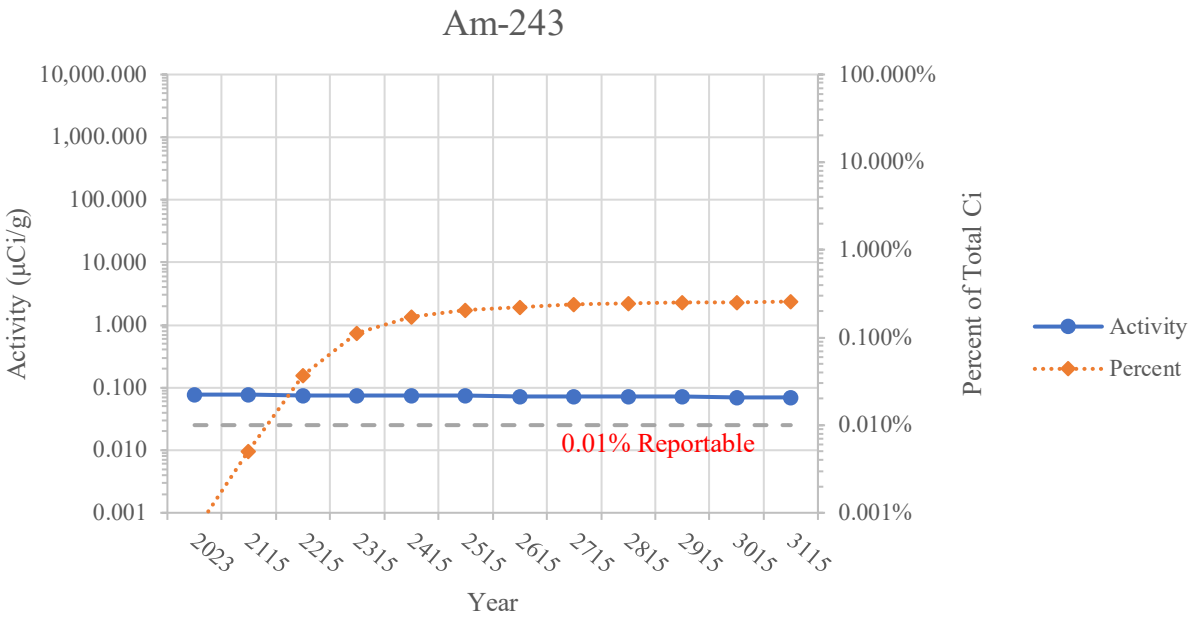
23. *ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. ANN. ICRP 38 (3).*

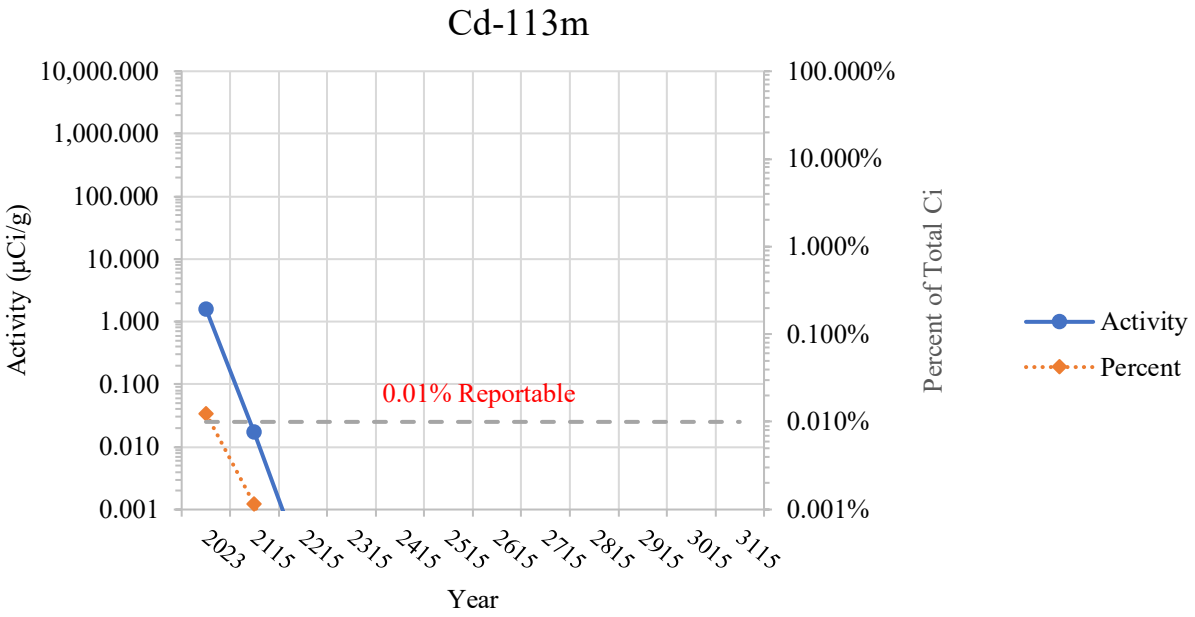
24. Hunter, S. C. *Software Test Plan for Radioactive Decay Calculations Performed by RadDecay® 7.00*; G-STP-A-00020, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2024.

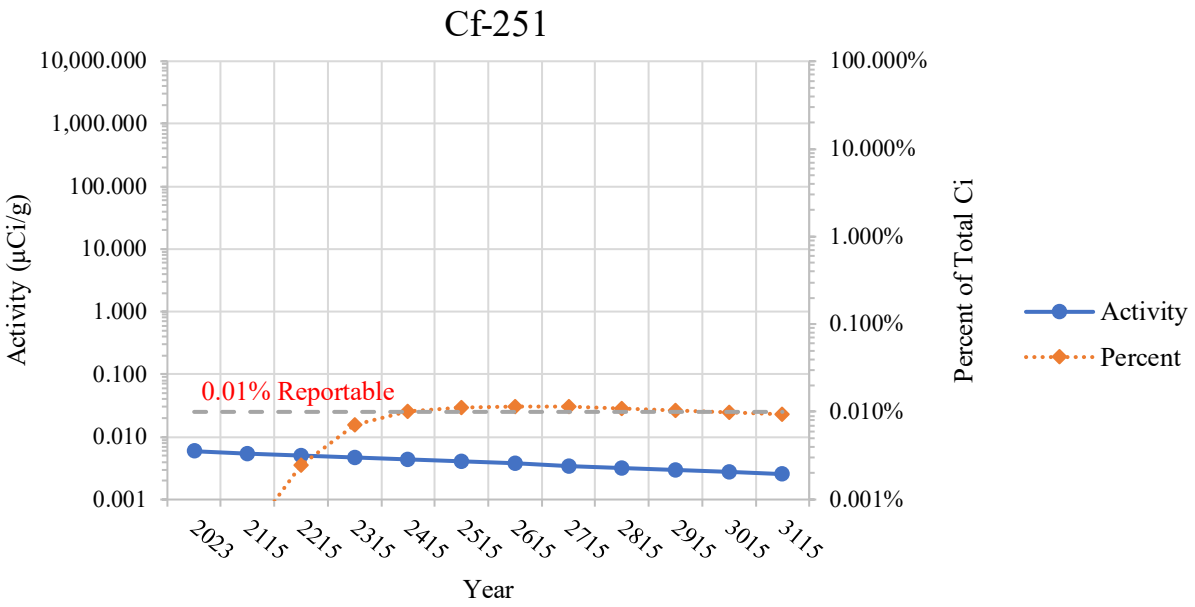
25. Hunter, S. C. *Software Test Results for Radioactive Decay Calculations Performed by RadDecay v. 7.00*; G-TR-W-00002, Rev. 0; Savannah River National Laboratory: Aiken, SC, 2024.

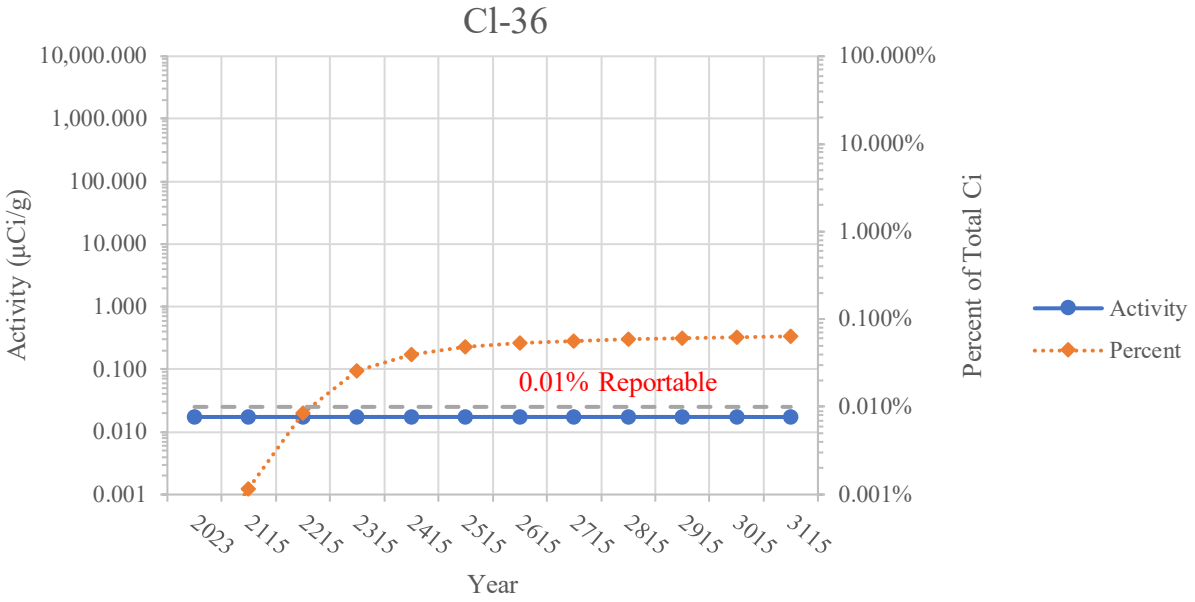
Appendix A. Individual Radionuclide Concentrations

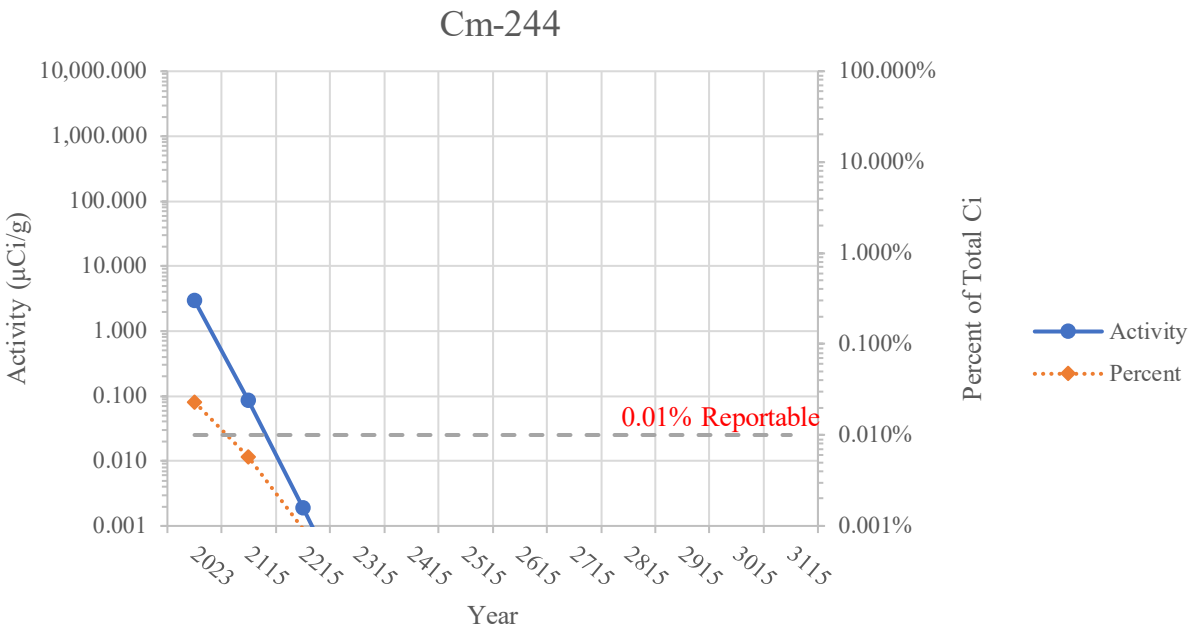


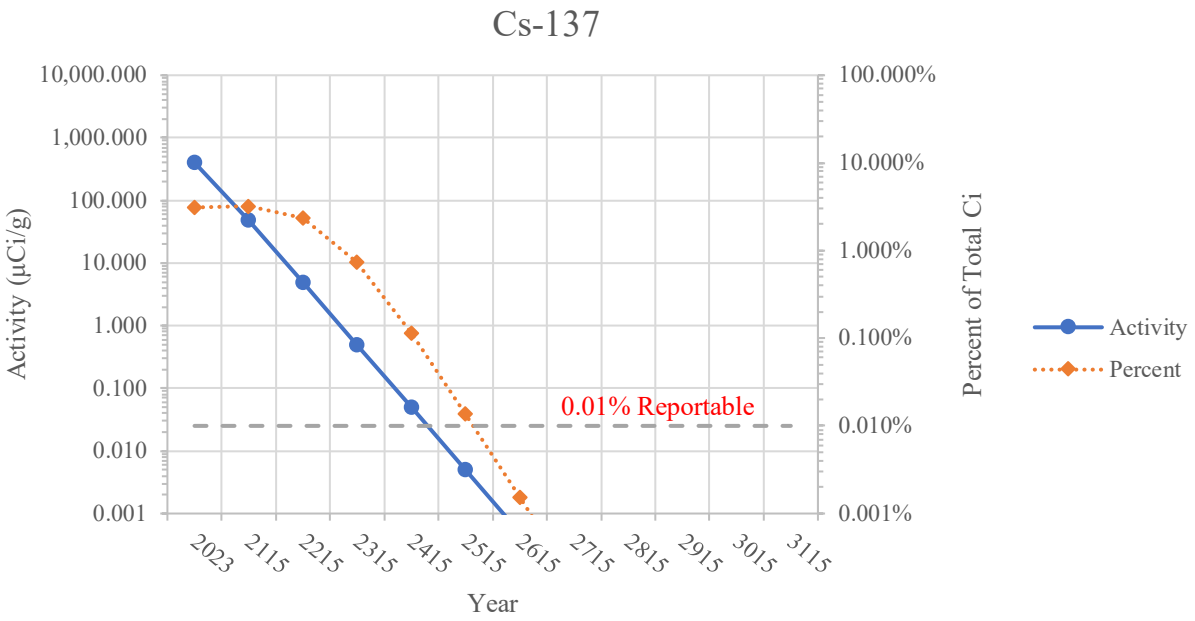


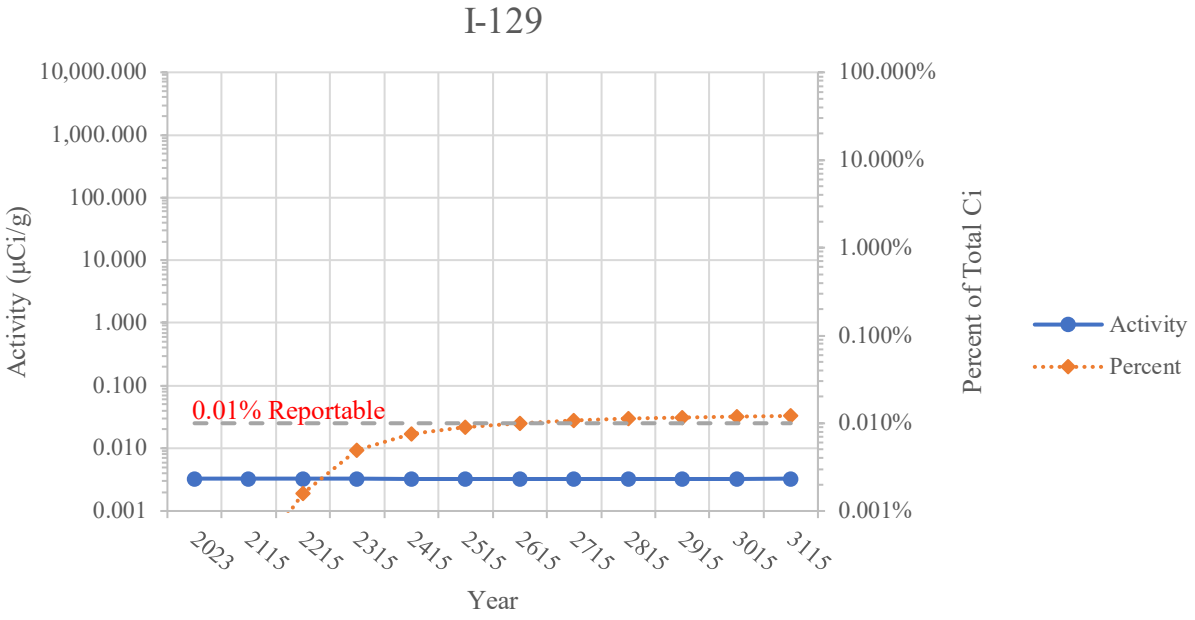

Cd-113m
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

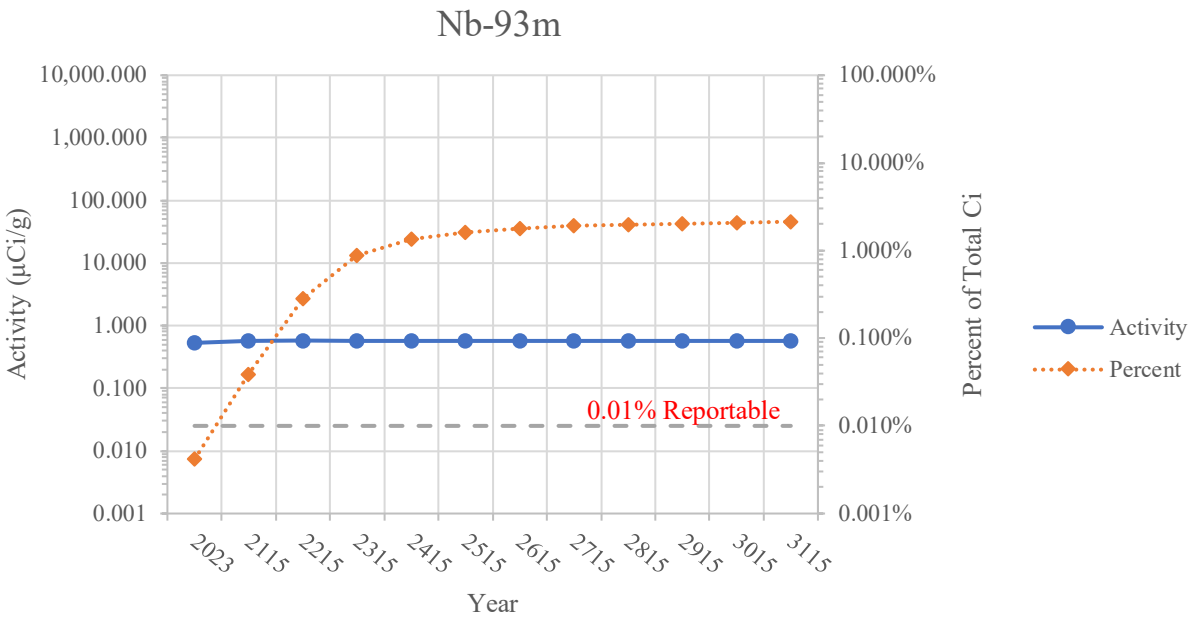

Cf-251
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

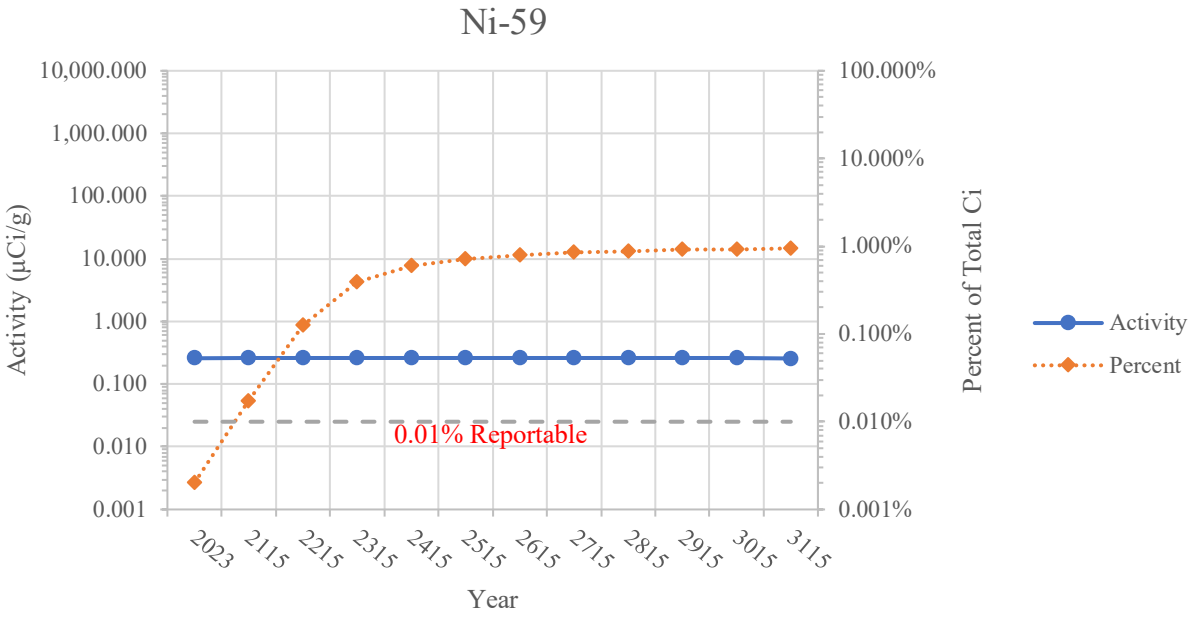

Cl-36
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

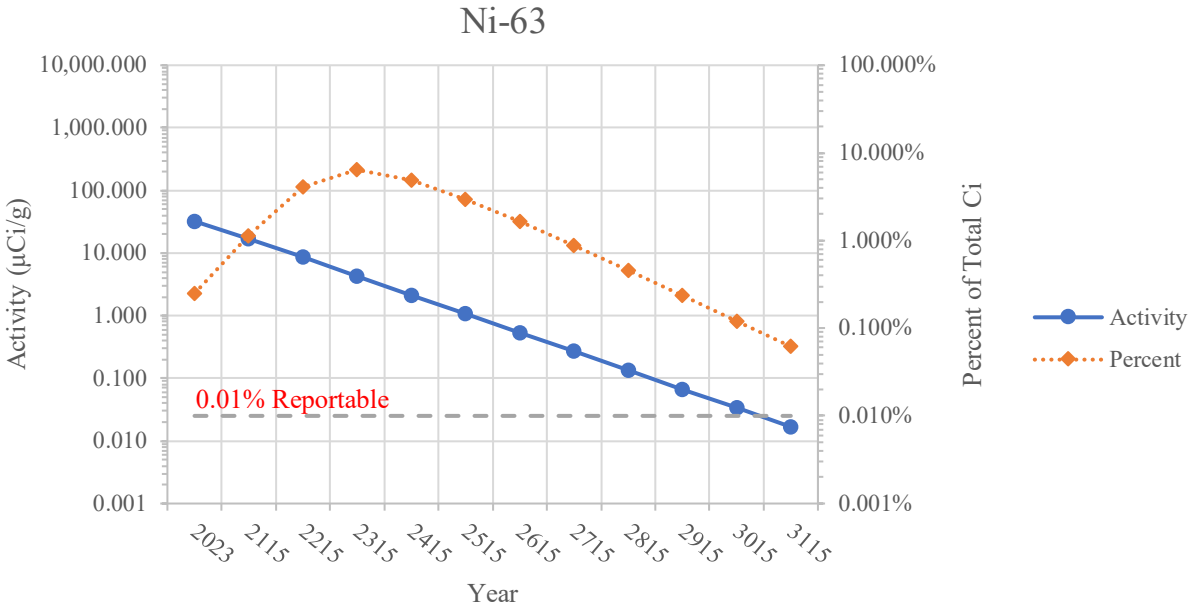

Cm-244
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

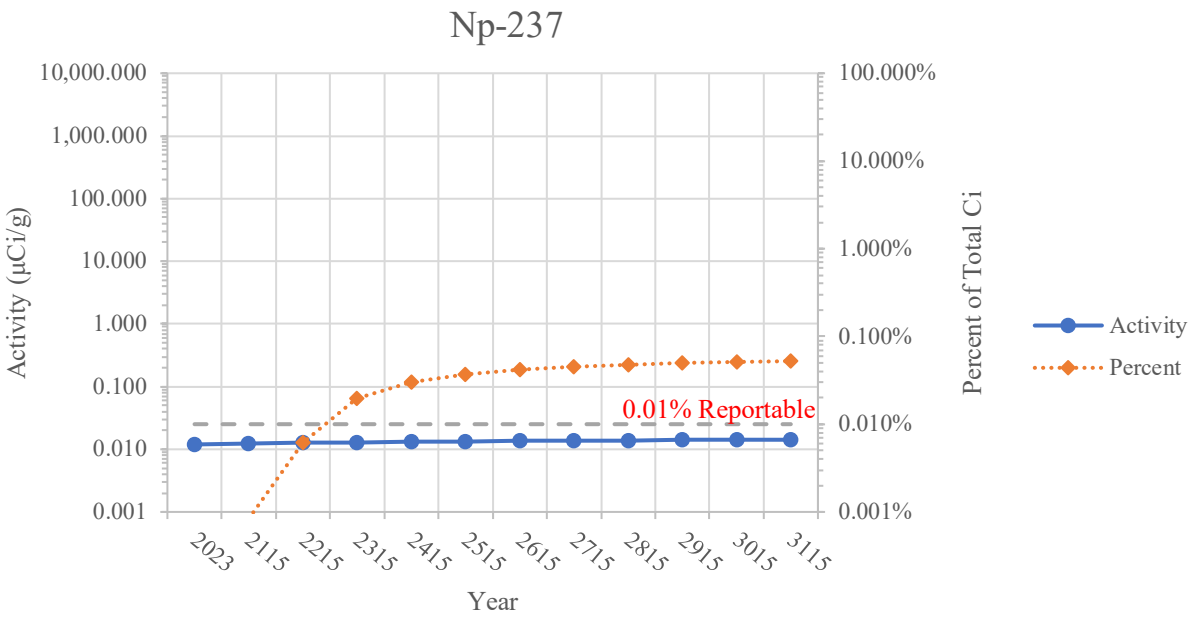

Cs-137
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

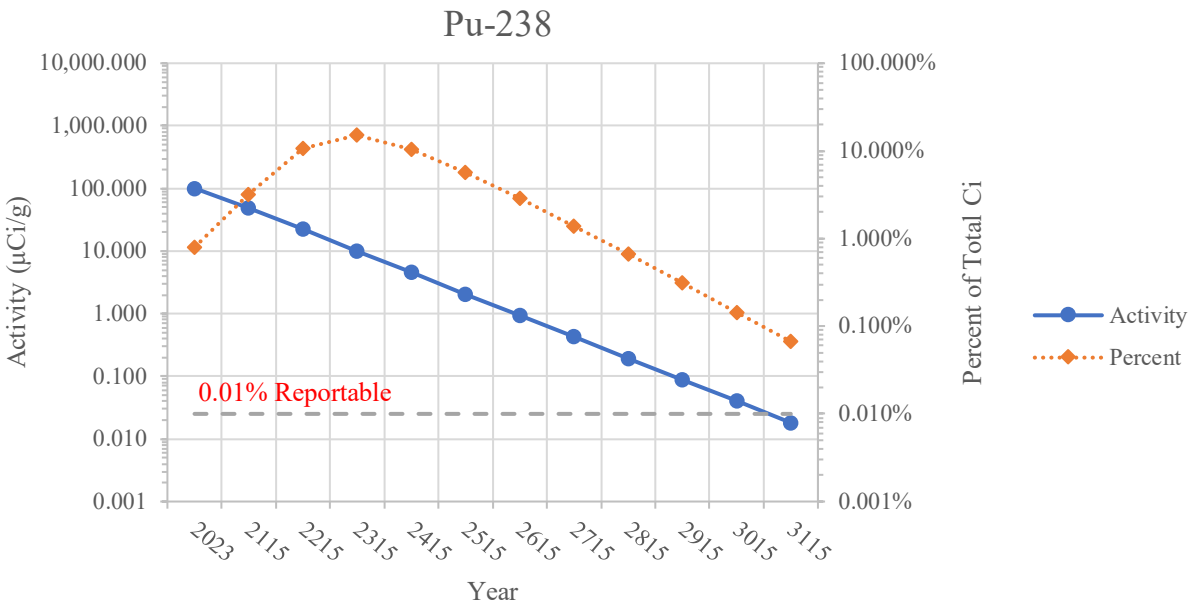

I-129
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

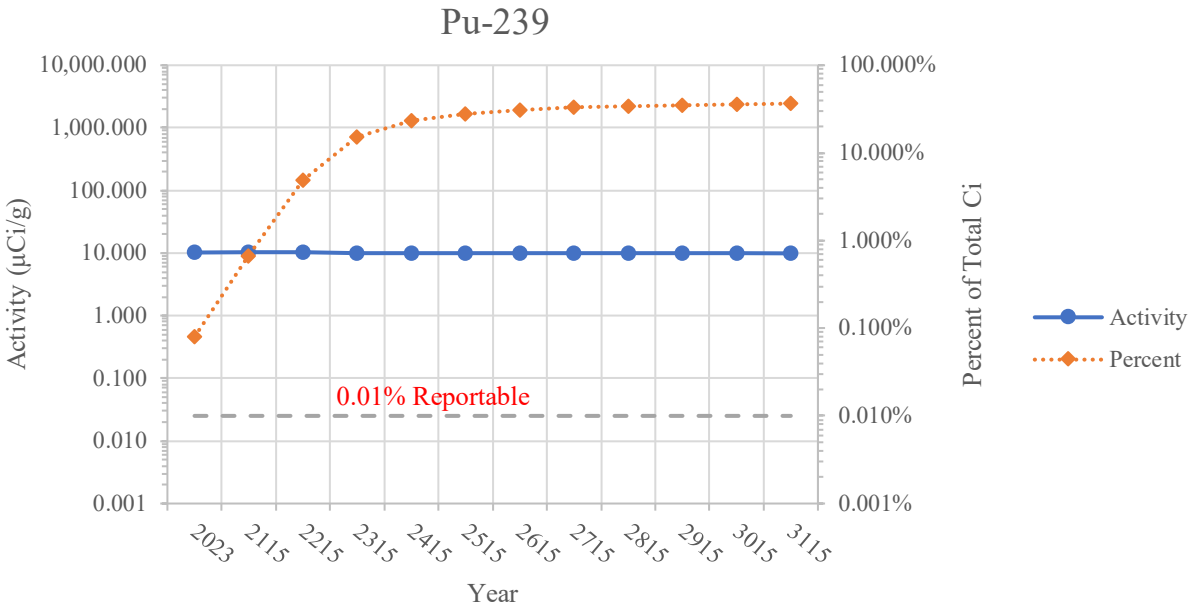

Nb-93m
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

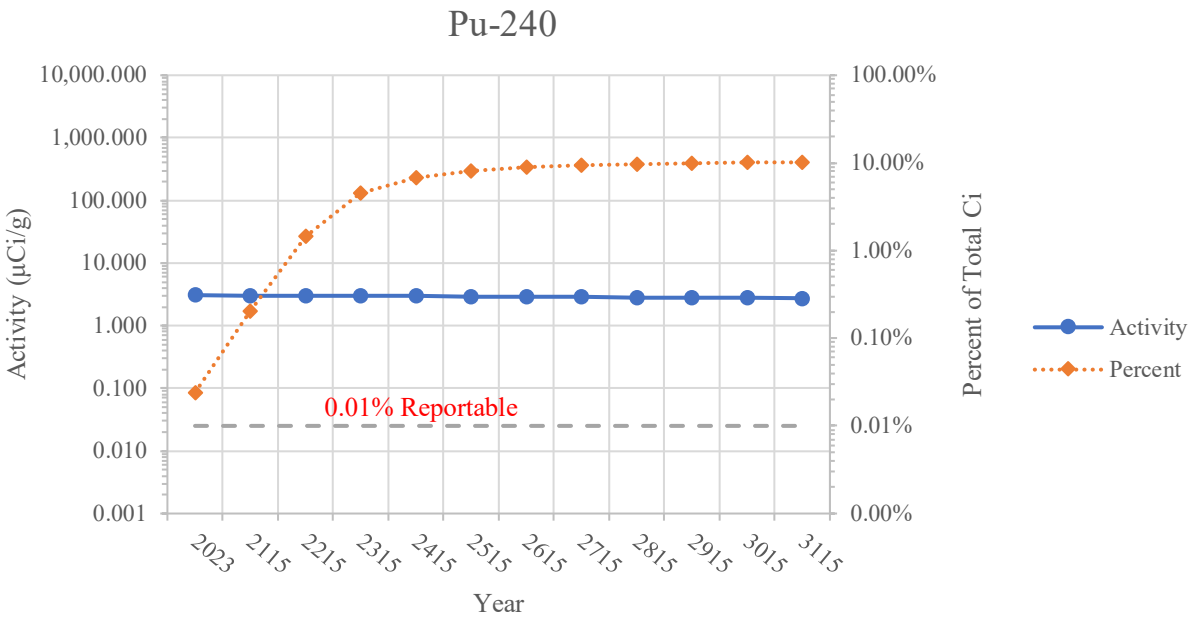

Ni-59
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

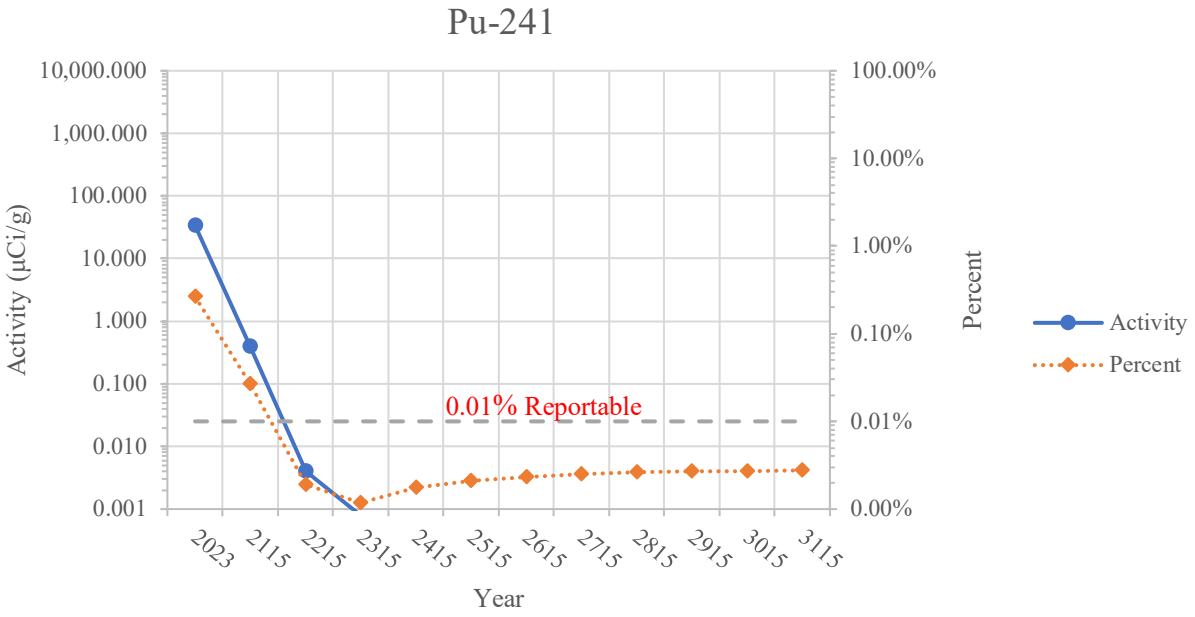

Ni-63
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

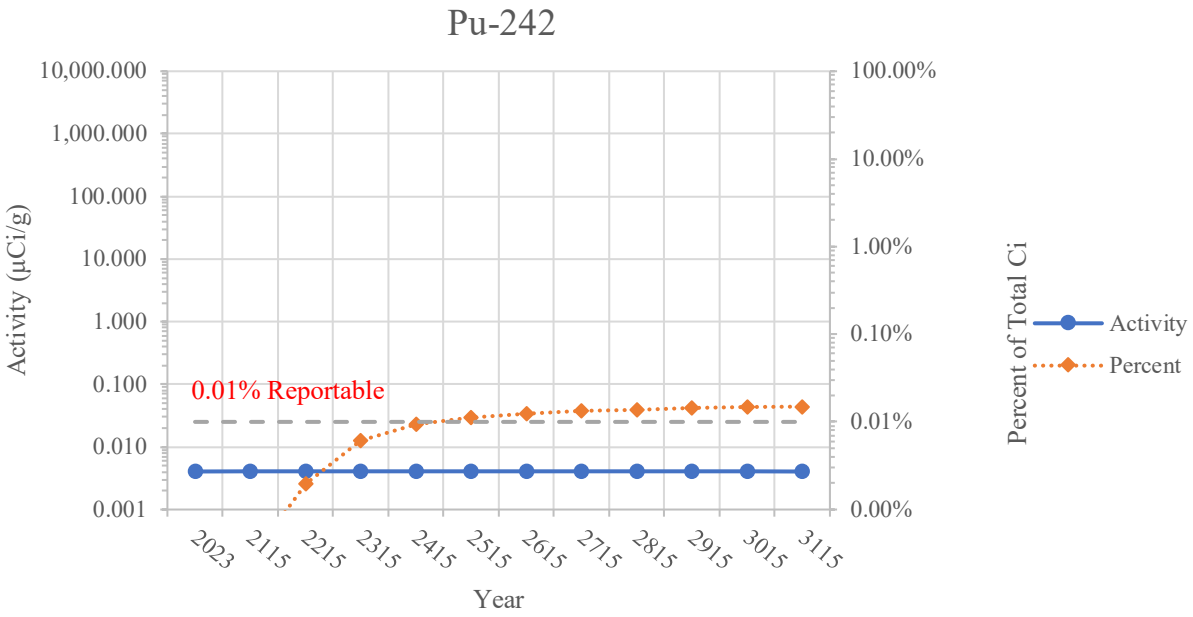

Np-237
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

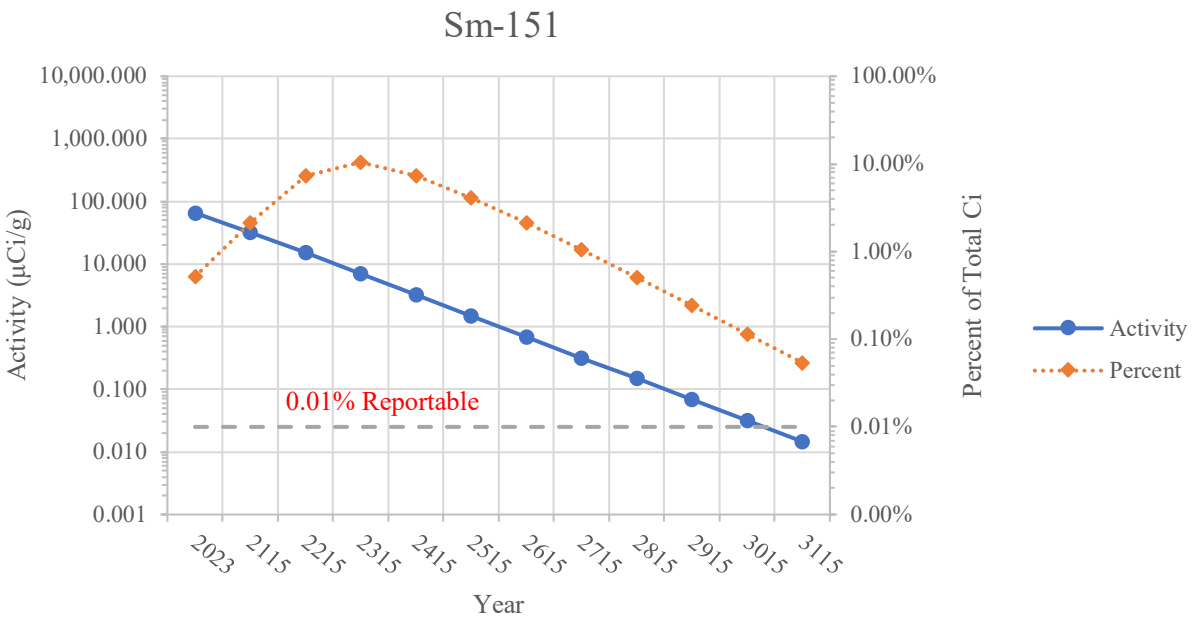

Pu-238
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

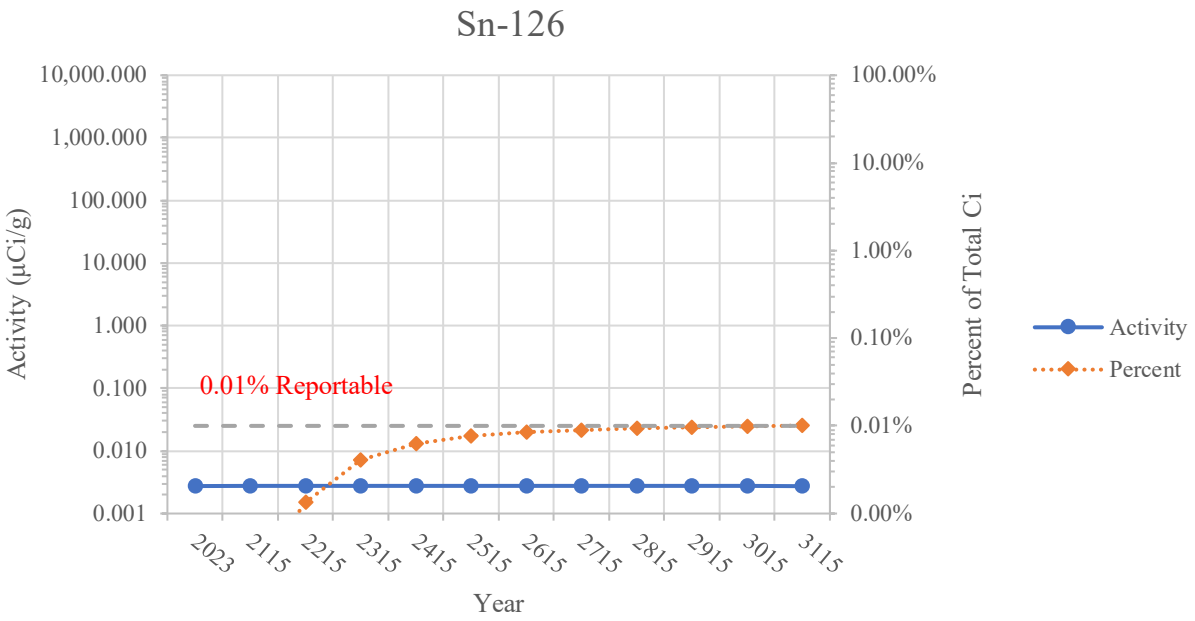

Pu-239
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

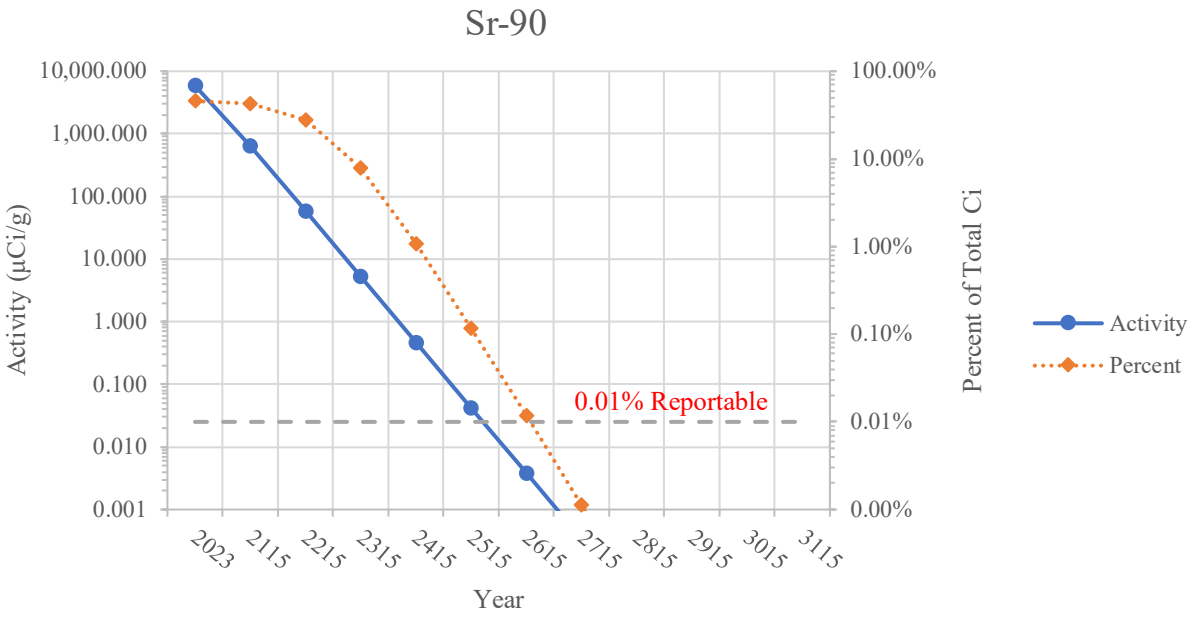

Pu-240
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

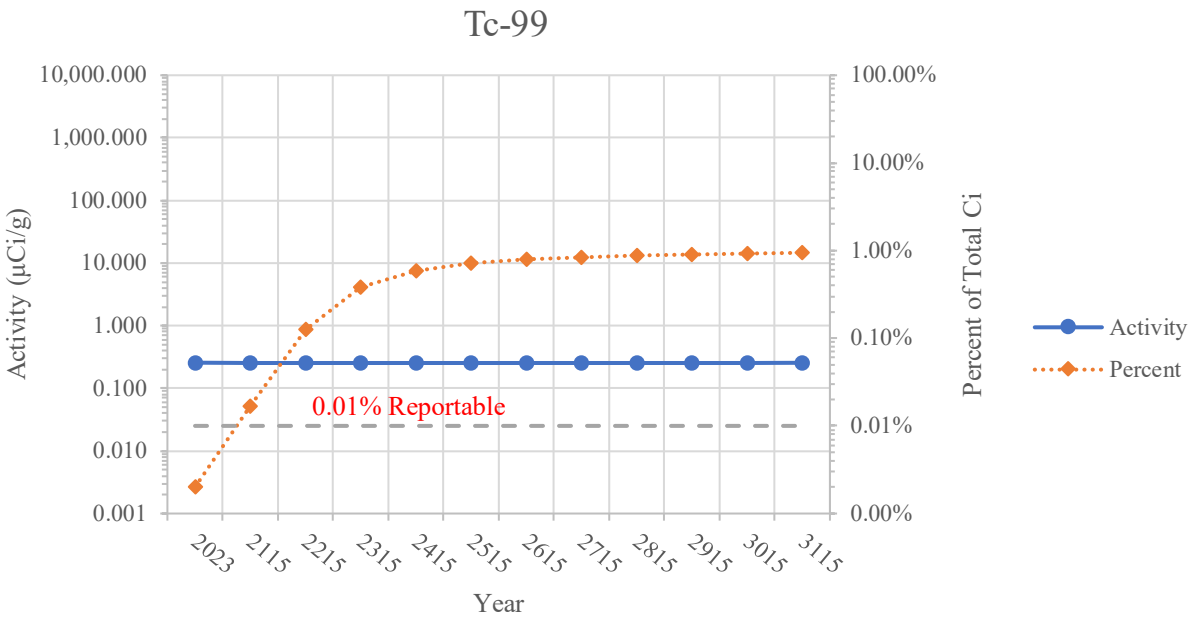

Pu-241
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

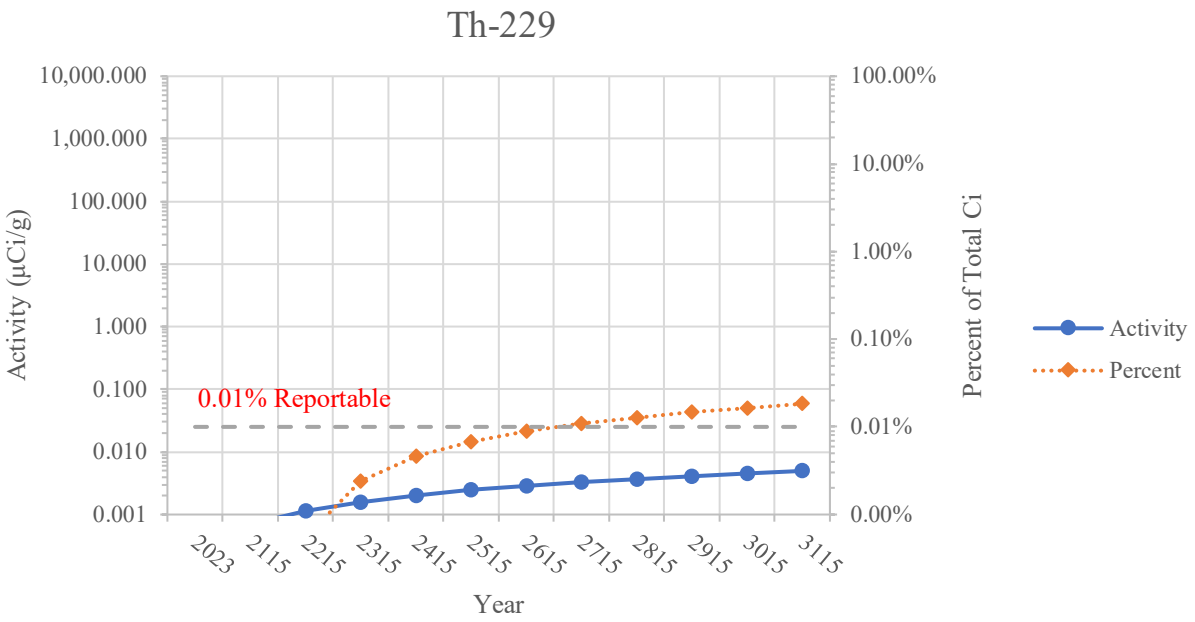

Pu-242
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

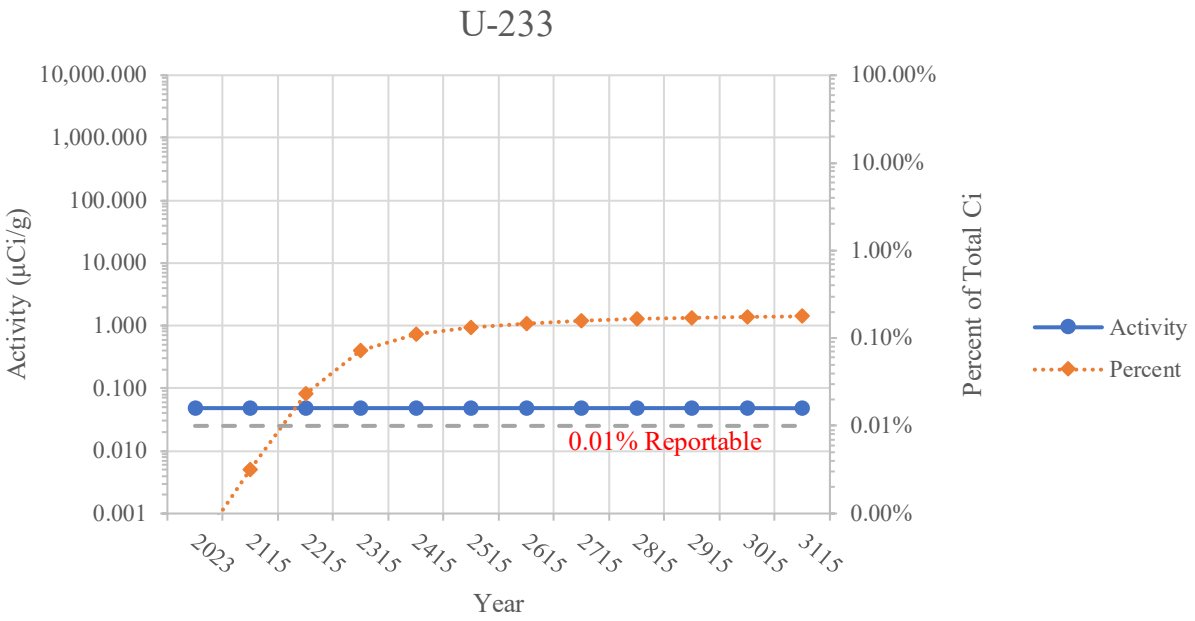

Sm-151
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

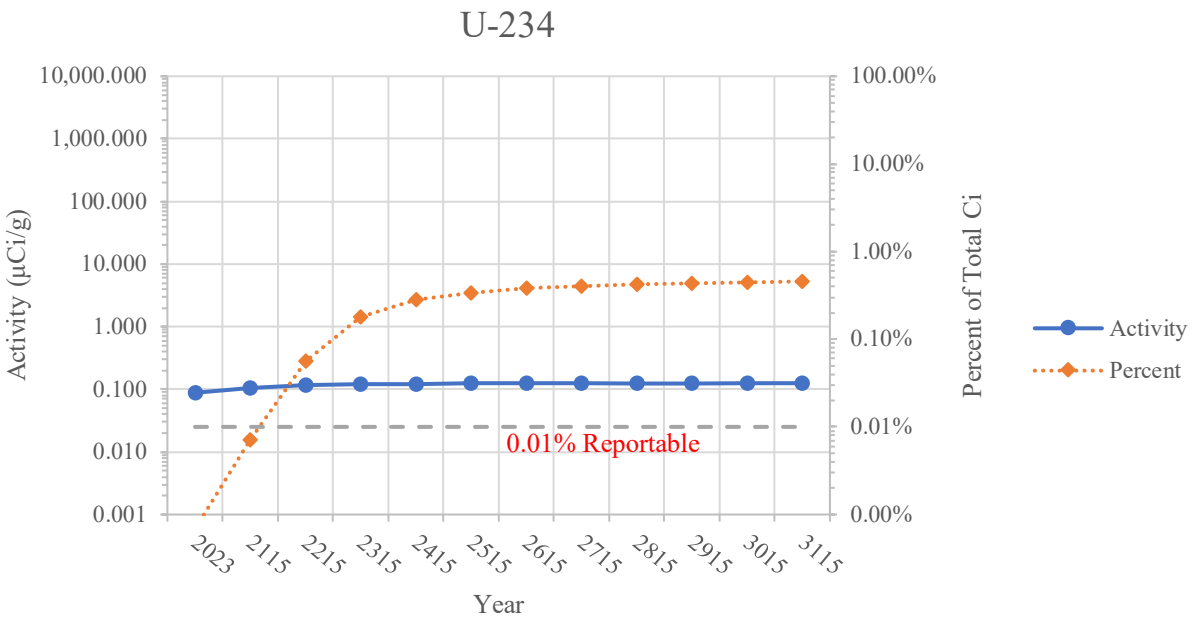

Sn-126
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

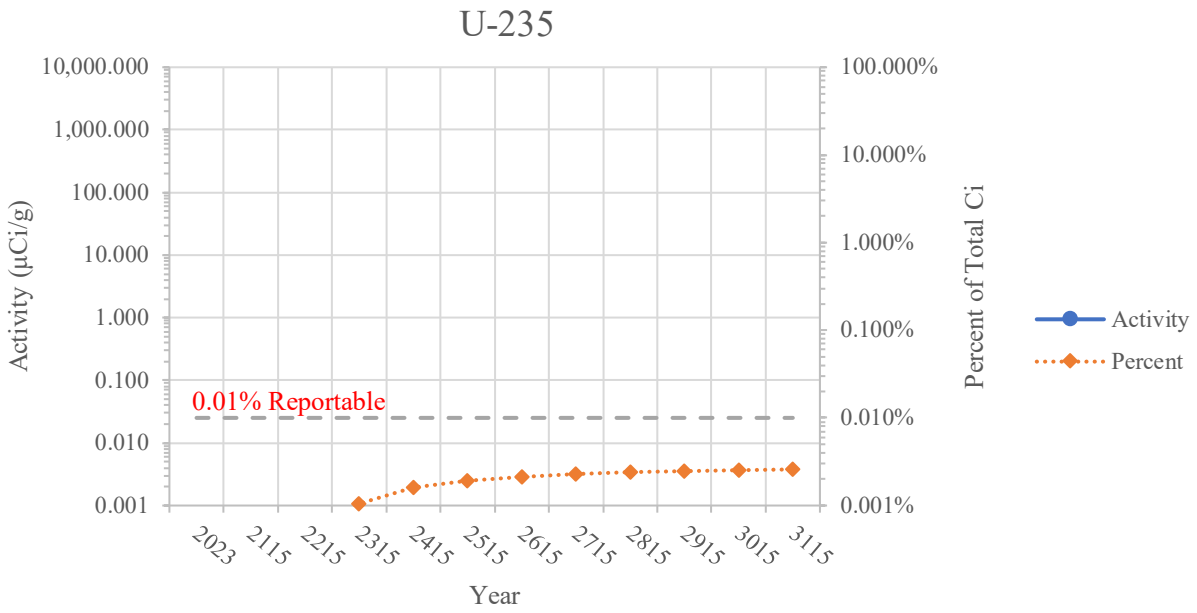

Sr-90
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

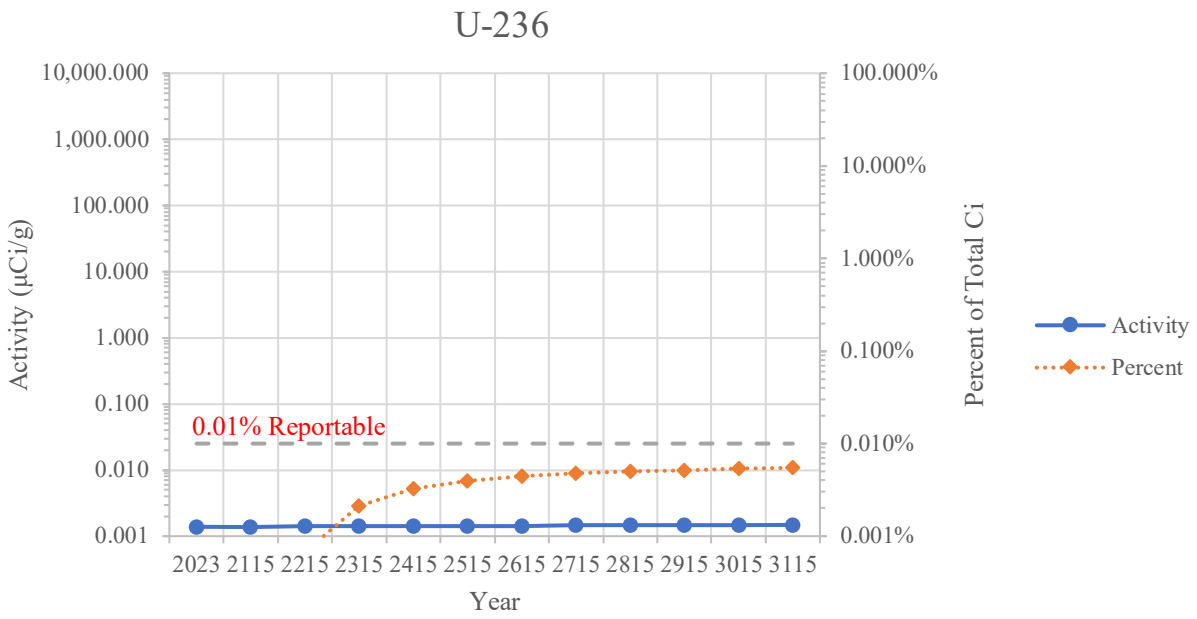

Tc-99
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

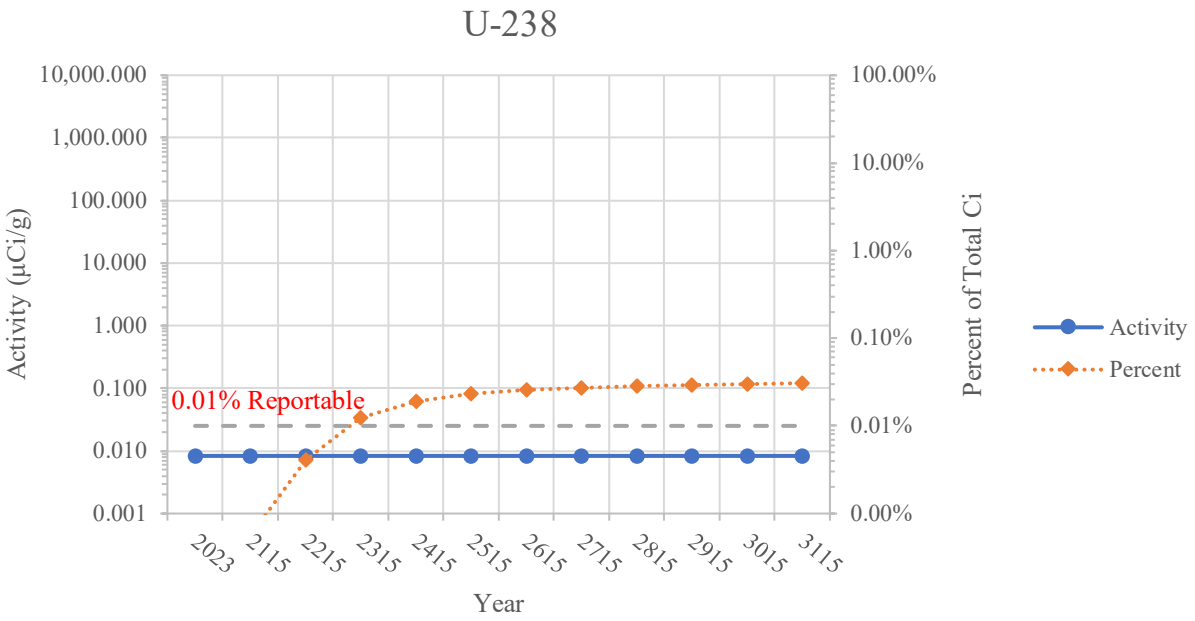

Th-229
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

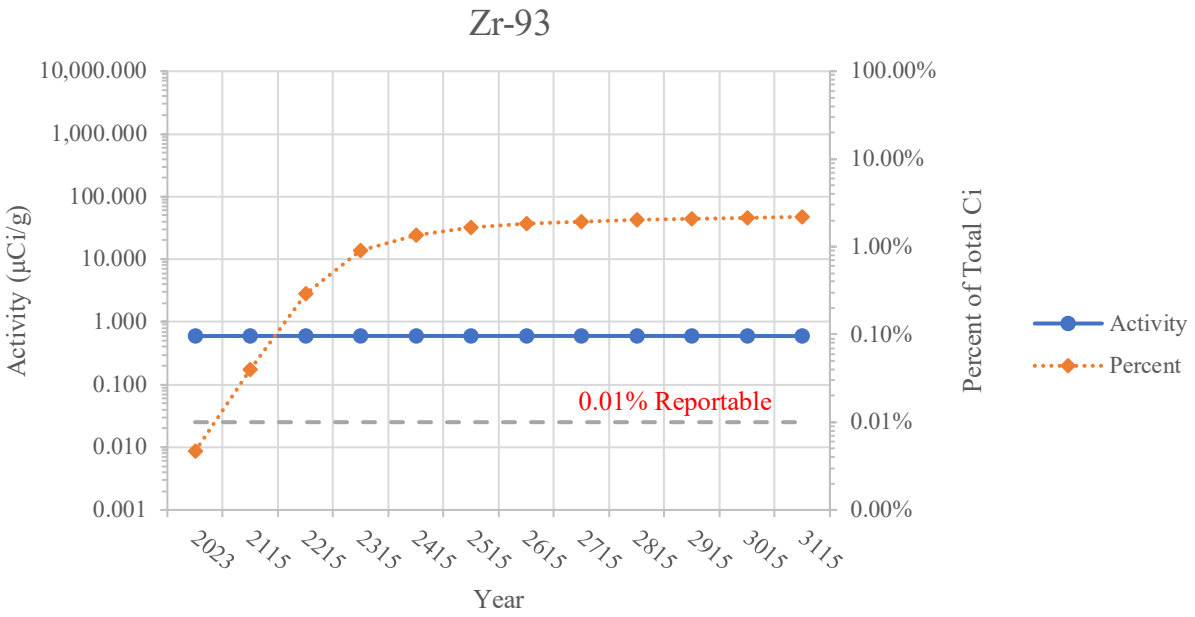

U-233
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent


U-234
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent


U-235
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent


U-236
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.000%
10.000%
1.000%
0.100%
0.010%
0.001%
Percent of Total Ci
0.01% Reportable
2023 2115 2215 2315 2415 2515 2615 2715 2815 2915 3015 3115
Year
Activity
Percent


U-238
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent


Zr-93
10,000.000
1,000.000
100.000
10.000
1.000
0.100
0.010
0.001
Activity (µCi/g)
100.00%
10.00%
1.00%
0.10%
0.01%
0.00%
Percent of Total Ci
0.01% Reportable
2023
2115
2215
2315
2415
2515
2615
2715
2815
2915
3015
3115
Year
Activity
Percent

Appendix B. Reportable Radionuclides for Sludge Batches 1b to 10. Red Shaded Cells Indicate Reportable Radionuclide.

Radionuclide	SB1b	SB2	SB3	SB4	SB5	SB6	SB7a	SB7b	SB8	SB9	SB10
C-14											
Cl-36											
Ni-59											
Ni-63											
Se-79											
Sr-90											
Zr-93											
Nb-93m											
Tc-99											
Cd-113m											
Sn-121m											
Sn-126											
I-129											
Cs-135											
Cs-137											
Sm-151											
Ac-227											
Th-229											
Pa-231											
U-233											
U-234											
Np-237											
U-238											
Pu-238											
Pu-239											
Pu-240											
Am-241											
Pu-241											
Pu-242											
Am-242m											
Am-243											
Cm-244											
Cm-245											
Cm-246											
Bk-247											
Cm-247											
Cm-248											
Cf-249											
Cf-251											

Appendix C. RadDecay® v.7.00 Output Activities for the Specified Index Years

Index year:		2023	2115	2215	2315	2415	2515	2615	2715	2815	2915	3015	3115
Name	Half-Life (y)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)
Ac-225	2.74E-02	0.00E+00	6.93E-04	1.14E-03	1.58E-03	2.02E-03	2.45E-03	2.88E-03	3.31E-03	3.73E-03	4.15E-03	4.56E-03	4.97E-03
Ac-227	2.18E+01	1.36E-03	4.77E-04	4.30E-04	4.28E-04	4.29E-04	4.29E-04	4.30E-04	4.30E-04	4.31E-04	4.31E-04	4.32E-04	4.33E-04
Ac-228	7.02E-04	0.00E+00	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Am-241	4.32E+02	1.24E+01	1.17E+01	9.99E+00	8.51E+00	7.25E+00	6.17E+00	5.26E+00	4.48E+00	3.82E+00	3.25E+00	2.77E+00	2.36E+00
Am-242	1.83E-03	0.00E+00	7.80E-03	4.77E-03	2.92E-03	1.79E-03	1.09E-03	6.68E-04	4.09E-04	2.50E-04	1.53E-04	9.35E-05	5.72E-05
Am-242m	1.41E+02	1.23E-02	7.84E-03	4.79E-03	2.93E-03	1.79E-03	1.10E-03	6.71E-04	4.10E-04	2.51E-04	1.54E-04	9.39E-05	5.74E-05
Am-243	7.37E+03	7.72E-02	7.65E-02	7.58E-02	7.51E-02	7.44E-02	7.37E-02	7.30E-02	7.23E-02	7.17E-02	7.10E-02	7.03E-02	6.97E-02
At-217	1.02E-09	0.00E+00	6.93E-04	1.14E-03	1.58E-03	2.02E-03	2.45E-03	2.88E-03	3.31E-03	3.73E-03	4.15E-03	4.56E-03	4.97E-03
At-218	4.76E-08	0.00E+00	3.37E-09	6.48E-09	1.04E-08	1.50E-08	2.04E-08	2.65E-08	3.33E-08	4.08E-08	4.89E-08	5.77E-08	6.70E-08
At-219	1.78E-06	0.00E+00	3.95E-10	3.56E-10	3.54E-10	3.55E-10	3.55E-10	3.56E-10	3.56E-10	3.57E-10	3.57E-10	3.58E-10	3.58E-10
Ba-133	1.05E+01	5.10E-02	1.19E-04	1.64E-07	2.25E-10	3.09E-13	4.26E-16	5.86E-19	8.06E-22	1.11E-24	1.52E-27	2.10E-30	2.88E-33
Ba-137m	4.86E-06	3.81E+02	4.59E+01	4.61E+00	4.64E-01	4.66E-02	4.68E-03	4.70E-04	4.73E-05	4.75E-06	4.77E-07	4.80E-08	4.82E-09
Bi-210	1.37E-02	0.00E+00	1.28E-05	2.72E-05	4.54E-05	6.73E-05	9.31E-05	1.23E-04	1.56E-04	1.92E-04	2.32E-04	2.74E-04	3.20E-04
Bi-211	4.07E-06	0.00E+00	4.77E-04	4.30E-04	4.28E-04	4.29E-04	4.29E-04	4.30E-04	4.30E-04	4.31E-04	4.31E-04	4.32E-04	4.33E-04
Bi-212	1.15E-04	0.00E+00	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Bi-213	8.67E-05	0.00E+00	6.93E-04	1.14E-03	1.58E-03	2.02E-03	2.45E-03	2.88E-03	3.31E-03	3.73E-03	4.15E-03	4.56E-03	4.97E-03
Bi-214	3.79E-05	0.00E+00	1.69E-05	3.24E-05	5.18E-05	7.50E-05	1.02E-04	1.33E-04	1.67E-04	2.04E-04	2.45E-04	2.88E-04	3.35E-04
Bi-215	1.45E-05	0.00E+00	3.83E-10	3.45E-10	3.44E-10	3.44E-10	3.45E-10	3.45E-10	3.46E-10	3.46E-10	3.46E-10	3.47E-10	3.47E-10
Bk-247	1.38E+03	1.33E-04	1.27E-04	1.21E-04	1.15E-04	1.09E-04	1.04E-04	9.86E-05	9.38E-05	8.92E-05	8.48E-05	8.07E-05	7.67E-05
Cd-113	7.70E+15	0.00E+00	4.07E-18	4.11E-18	4.11E-18	4.11E-18	4.11E-18	4.11E-18	4.11E-18	4.11E-18	4.11E-18	4.11E-18	4.11E-18
Cd-113m	1.41E+01	1.61E+00	1.74E-02	1.28E-04	9.36E-07	6.86E-09	5.03E-11	3.69E-13	2.70E-15	1.98E-17	1.45E-19	1.06E-21	7.79E-24
Ce-144	7.80E-01	5.16E-01	1.62E-36	4.14E-75	1.06E-113	2.72E-152	6.95E-191	1.78E-229	4.56E-268	1.17E-306	0.00E+00	0.00E+00	0.00E+00
Cf-249	3.51E+02	2.19E-03	1.82E-03	1.50E-03	1.23E-03	1.01E-03	8.28E-04	6.79E-04	5.58E-04	4.58E-04	3.76E-04	3.08E-04	2.53E-04
Cf-250	1.31E+01	1.02E-02	7.78E-05	3.88E-07	1.94E-09	9.69E-12	4.84E-14	2.42E-16	1.21E-18	6.03E-21	3.01E-23	1.51E-25	7.52E-28
Cf-251	9.00E+02	5.91E-03	5.50E-03	5.09E-03	4.72E-03	4.37E-03	4.04E-03	3.74E-03	3.47E-03	3.21E-03	2.97E-03	2.75E-03	2.55E-03
Cf-252	2.65E+00	1.02E-02	3.45E-13	1.43E-24	5.96E-36	2.48E-47	1.03E-58	4.29E-70	1.78E-81	7.41E-93	3.08E-104	1.28E-115	5.33E-127

Index year:		2023	2115	2215	2315	2415	2515	2615	2715	2815	2915	3015	3115
Name	Half-Life (y)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)
Cl-36	3.01E+05	1.73E-02	1.73E-02	1.73E-02	1.73E-02	1.73E-02	1.73E-02	1.73E-02	1.73E-02	1.73E-02	1.73E-02	1.73E-02	1.73E-02
Cm-242	4.46E-01	1.02E-02	6.47E-03	3.96E-03	2.42E-03	1.48E-03	9.06E-04	5.54E-04	3.39E-04	2.07E-04	1.27E-04	7.76E-05	4.74E-05
Cm-243	2.91E+01	5.49E-02	6.13E-03	5.67E-04	5.23E-05	4.83E-06	4.47E-07	4.13E-08	3.81E-09	3.52E-10	3.25E-11	3.00E-12	2.77E-13
Cm-244	1.81E+01	2.96E+00	8.73E-02	1.90E-03	4.12E-05	8.94E-07	1.94E-08	4.22E-10	9.16E-12	1.99E-13	4.32E-15	9.39E-17	2.04E-18
Cm-245	8.50E+03	7.37E-04	7.46E-04	7.54E-04	7.59E-04	7.61E-04	7.63E-04	7.63E-04	7.61E-04	7.59E-04	7.57E-04	7.53E-04	7.49E-04
Cm-246	4.76E+03	1.92E-03	1.93E-03	1.90E-03	1.87E-03	1.84E-03	1.82E-03	1.79E-03	1.76E-03	1.74E-03	1.71E-03	1.69E-03	1.66E-03
Cm-247	1.56E+07	1.19E-08	3.52E-08	5.87E-08	8.05E-08	1.01E-07	1.19E-07	1.37E-07	1.53E-07	1.68E-07	1.81E-07	1.94E-07	2.06E-07
Cm-248	3.48E+05	4.80E-06	4.88E-06	4.88E-06	4.88E-06	4.87E-06	4.87E-06	4.87E-06	4.87E-06	4.87E-06	4.87E-06	4.87E-06	4.87E-06
Co-60	5.27E+00	9.54E-02	5.32E-07	1.03E-12	2.01E-18	3.92E-24	7.63E-30	1.49E-35	2.89E-41	5.63E-47	1.10E-52	2.13E-58	4.15E-64
Cs-134	2.06E+00	1.11E+00	4.28E-14	1.13E-28	2.97E-43	7.83E-58	2.06E-72	5.44E-87	1.43E-101	3.78E-116	9.96E-131	2.62E-145	6.92E-160
Cs-135	2.30E+06	2.51E-03	2.51E-03	2.51E-03	2.51E-03	2.51E-03	2.51E-03	2.51E-03	2.51E-03	2.51E-03	2.51E-03	2.51E-03	2.51E-03
Cs-137	3.02E+01	4.03E+02	4.86E+01	4.89E+00	4.91E-01	4.94E-02	4.96E-03	4.98E-04	5.01E-05	5.03E-06	5.06E-07	5.08E-08	5.11E-09
Eu-154	8.59E+00	3.90E+00	2.33E-03	7.33E-07	2.30E-10	7.22E-14	2.27E-17	7.11E-21	2.23E-24	7.01E-28	2.20E-31	6.91E-35	2.17E-38
Eu-155	4.76E+00	1.98E-01	3.02E-07	1.44E-13	6.83E-20	3.25E-26	1.55E-32	7.35E-39	3.50E-45	1.66E-51	7.91E-58	3.76E-64	1.79E-70
Fr-221	9.32E-06	0.00E+00	6.93E-04	1.14E-03	1.58E-03	2.02E-03	2.45E-03	2.88E-03	3.31E-03	3.73E-03	4.15E-03	4.56E-03	4.97E-03
Fr-223	4.19E-05	0.00E+00	6.58E-06	5.93E-06	5.91E-06	5.91E-06	5.92E-06	5.93E-06	5.94E-06	5.95E-06	5.95E-06	5.96E-06	5.97E-06
Hg-206	1.55E-05	0.00E+00	2.44E-13	5.17E-13	8.62E-13	1.28E-12	1.77E-12	2.33E-12	2.96E-12	3.65E-12	4.40E-12	5.21E-12	6.08E-12
I-129	1.57E+07	3.29E-03	3.29E-03	3.29E-03	3.29E-03	3.29E-03	3.29E-03	3.29E-03	3.29E-03	3.29E-03	3.29E-03	3.29E-03	3.29E-03
Nb-93m	1.61E+01	5.31E-01	5.79E-01	5.80E-01	5.80E-01	5.80E-01	5.80E-01	5.80E-01	5.80E-01	5.80E-01	5.80E-01	5.80E-01	5.80E-01
Nd-144	2.29E+15	0.00E+00	1.76E-16	1.76E-16	1.76E-16	1.76E-16	1.76E-16	1.76E-16	1.76E-16	1.76E-16	1.76E-16	1.76E-16	1.76E-16
Ni-59	1.01E+05	2.59E-01	2.58E-01	2.58E-01	2.58E-01	2.58E-01	2.58E-01	2.58E-01	2.57E-01	2.57E-01	2.57E-01	2.57E-01	2.57E-01
Ni-63	1.00E+02	3.22E+01	1.71E+01	8.53E+00	4.27E+00	2.14E+00	1.07E+00	5.35E-01	2.68E-01	1.34E-01	6.70E-02	3.35E-02	1.68E-02
Np-237	2.14E+06	1.19E-02	1.23E-02	1.26E-02	1.29E-02	1.32E-02	1.34E-02	1.36E-02	1.38E-02	1.39E-02	1.40E-02	1.41E-02	1.42E-02
Np-238	5.80E-03	0.00E+00	3.53E-05	2.16E-05	1.32E-05	8.07E-06	4.94E-06	3.02E-06	1.85E-06	1.13E-06	6.91E-07	4.23E-07	2.59E-07
Np-239	6.46E-03	0.00E+00	7.65E-02	7.58E-02	7.51E-02	7.44E-02	7.37E-02	7.30E-02	7.23E-02	7.17E-02	7.10E-02	7.03E-02	6.97E-02
Np-240	1.18E-04	0.00E+00	3.91E-15	8.16E-15	1.24E-14	1.67E-14	2.09E-14	2.52E-14	2.94E-14	3.37E-14	3.79E-14	4.22E-14	4.64E-14
Np-240m	1.37E-05	0.00E+00	3.56E-12	7.42E-12	1.13E-11	1.52E-11	1.90E-11	2.29E-11	2.67E-11	3.06E-11	3.45E-11	3.83E-11	4.22E-11
Pa-231	3.28E+04	4.27E-04	4.27E-04	4.28E-04	4.28E-04	4.29E-04	4.29E-04	4.30E-04	4.30E-04	4.31E-04	4.32E-04	4.32E-04	4.33E-04

Index year:		2023	2115	2215	2315	2415	2515	2615	2715	2815	2915	3015	3115
Name	Half-Life (y)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)
Pa-233	7.39E-02	0.00E+00	1.23E-02	1.26E-02	1.29E-02	1.32E-02	1.34E-02	1.36E-02	1.38E-02	1.39E-02	1.40E-02	1.41E-02	1.42E-02
Pa-234	7.65E-04	0.00E+00	1.33E-05	1.33E-05	1.33E-05	1.33E-05	1.33E-05	1.33E-05	1.33E-05	1.33E-05	1.33E-05	1.33E-05	1.33E-05
Pa-234m	2.23E-06	0.00E+00	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03
Pb-209	3.71E-04	0.00E+00	6.93E-04	1.14E-03	1.58E-03	2.02E-03	2.45E-03	2.88E-03	3.31E-03	3.73E-03	4.15E-03	4.56E-03	4.97E-03
Pb-210	2.22E+01	3.06E-06	1.29E-05	2.72E-05	4.54E-05	6.74E-05	9.31E-05	1.23E-04	1.56E-04	1.92E-04	2.32E-04	2.74E-04	3.20E-04
Pb-211	6.87E-05	0.00E+00	4.77E-04	4.30E-04	4.28E-04	4.29E-04	4.29E-04	4.30E-04	4.30E-04	4.31E-04	4.31E-04	4.32E-04	4.33E-04
Pb-212	1.22E-03	0.00E+00	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Pb-214	5.10E-05	0.00E+00	1.69E-05	3.24E-05	5.18E-05	7.50E-05	1.02E-04	1.33E-04	1.67E-04	2.04E-04	2.45E-04	2.88E-04	3.35E-04
Pd-107	6.50E+06	4.68E-04	4.68E-04	4.68E-04	4.68E-04	4.68E-04	4.68E-04	4.68E-04	4.68E-04	4.68E-04	4.68E-04	4.68E-04	4.68E-04
Pm-147	2.62E+00	3.97E+01	1.10E-09	3.69E-21	1.24E-32	4.14E-44	1.39E-55	4.65E-67	1.56E-78	5.22E-90	1.75E-101	5.87E-113	1.97E-124
Po-210	3.79E-01	0.00E+00	1.28E-05	2.71E-05	4.52E-05	6.72E-05	9.30E-05	1.22E-04	1.55E-04	1.92E-04	2.31E-04	2.74E-04	3.20E-04
Po-211	1.64E-08	0.00E+00	1.32E-06	1.19E-06	1.18E-06	1.18E-06	1.18E-06	1.19E-06	1.19E-06	1.19E-06	1.19E-06	1.19E-06	1.19E-06
Po-212	9.48E-15	0.00E+00	9.96E-04	9.96E-04	9.96E-04	9.96E-04	9.96E-04	9.96E-04	9.96E-04	9.96E-04	9.96E-04	9.96E-04	9.96E-04
Po-213	1.33E-13	0.00E+00	6.78E-04	1.12E-03	1.55E-03	1.98E-03	2.40E-03	2.82E-03	3.24E-03	3.65E-03	4.06E-03	4.47E-03	4.87E-03
Po-214	5.20E-12	0.00E+00	1.69E-05	3.24E-05	5.18E-05	7.50E-05	1.02E-04	1.33E-04	1.67E-04	2.04E-04	2.45E-04	2.88E-04	3.35E-04
Po-215	5.64E-11	0.00E+00	4.77E-04	4.30E-04	4.28E-04	4.29E-04	4.29E-04	4.30E-04	4.30E-04	4.31E-04	4.31E-04	4.32E-04	4.33E-04
Po-216	4.60E-09	0.00E+00	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Po-218	5.90E-06	0.00E+00	1.69E-05	3.24E-05	5.18E-05	7.50E-05	1.02E-04	1.33E-04	1.67E-04	2.04E-04	2.45E-04	2.88E-04	3.35E-04
Pr-144	3.29E-05	0.00E+00	1.62E-36	4.14E-75	1.06E-113	2.72E-152	6.95E-191	1.78E-229	4.56E-268	1.17E-306	0.00E+00	0.00E+00	0.00E+00
Pr-144m	1.37E-05	0.00E+00	1.58E-38	4.04E-77	1.04E-115	2.65E-154	6.79E-193	1.74E-231	4.46E-270	0.00E-01	0.00E+00	0.00E+00	0.00E+00
Pu-238	8.77E+01	1.01E+02	4.87E+01	2.21E+01	1.00E+01	4.55E+00	2.06E+00	9.37E-01	4.25E-01	1.93E-01	8.77E-02	3.98E-02	1.81E-02
Pu-239	2.41E+04	1.02E+01	1.02E+01	1.02E+01	1.01E+01	1.01E+01	1.01E+01	1.00E+01	1.00E+01	9.98E+00	9.95E+00	9.92E+00	9.90E+00
Pu-240	6.56E+03	3.08E+00	3.05E+00	3.02E+00	2.99E+00	2.96E+00	2.93E+00	2.90E+00	2.87E+00	2.84E+00	2.81E+00	2.78E+00	2.75E+00
Pu-241	1.44E+01	3.41E+01	4.01E-01	3.95E-03	7.83E-04	7.61E-04	7.63E-04	7.63E-04	7.62E-04	7.60E-04	7.57E-04	7.54E-04	7.50E-04
Pu-242	3.75E+05	4.05E-03	4.05E-03	4.05E-03	4.05E-03	4.05E-03	4.05E-03	4.05E-03	4.05E-03	4.05E-03	4.05E-03	4.05E-03	4.05E-03
Pu-243	5.67E-04	0.00E+00	3.52E-08	5.87E-08	8.05E-08	1.01E-07	1.19E-07	1.37E-07	1.53E-07	1.68E-07	1.81E-07	1.94E-07	2.06E-07
Pu-244	8.00E+07	0.00E+00	3.56E-12	7.43E-12	1.13E-11	1.52E-11	1.90E-11	2.29E-11	2.68E-11	3.06E-11	3.45E-11	3.84E-11	4.22E-11
Ra-223	3.13E-02	0.00E+00	4.77E-04	4.30E-04	4.28E-04	4.29E-04	4.29E-04	4.30E-04	4.30E-04	4.31E-04	4.31E-04	4.32E-04	4.33E-04

Index year:		2023	2115	2215	2315	2415	2515	2615	2715	2815	2915	3015	3115
Name	Half-Life (y)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)
Ra-224	1.00E-02	0.00E+00	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Ra-225	4.08E-02	0.00E+00	6.93E-04	1.14E-03	1.58E-03	2.02E-03	2.45E-03	2.88E-03	3.31E-03	3.73E-03	4.15E-03	4.56E-03	4.97E-03
Ra-226	1.60E+03	5.74E-06	1.69E-05	3.24E-05	5.18E-05	7.50E-05	1.02E-04	1.33E-04	1.67E-04	2.04E-04	2.45E-04	2.88E-04	3.35E-04
Ra-228	5.75E+00	0.00E+00	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Rh-106	9.45E-07	0.00E+00	1.03E-28	3.80E-58	1.41E-87	5.22E-117	1.94E-146	7.18E-176	2.66E-205	9.86E-235	3.66E-264	1.36E-293	0.00E+00
Rn-218	1.11E-09	0.00E+00	3.37E-12	6.48E-12	1.04E-11	1.50E-11	2.04E-11	2.65E-11	3.33E-11	4.08E-11	4.89E-11	5.77E-11	6.70E-11
Rn-219	1.26E-07	0.00E+00	4.77E-04	4.30E-04	4.28E-04	4.29E-04	4.29E-04	4.30E-04	4.30E-04	4.31E-04	4.31E-04	4.32E-04	4.33E-04
Rn-220	1.76E-06	0.00E+00	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Rn-222	1.05E-02	0.00E+00	1.69E-05	3.24E-05	5.18E-05	7.50E-05	1.02E-04	1.33E-04	1.67E-04	2.04E-04	2.45E-04	2.88E-04	3.35E-04
Ru-106	1.02E+00	1.22E-01	1.03E-28	3.80E-58	1.41E-87	5.22E-117	1.94E-146	7.18E-176	2.66E-205	9.86E-235	3.66E-264	1.36E-293	0.00E+00
Sb-125	2.76E+00	9.43E-02	8.61E-12	1.05E-22	1.29E-33	1.57E-44	1.93E-55	2.35E-66	2.88E-77	3.52E-88	4.31E-99	5.27E-110	6.44E-121
Sb-126	3.38E-02	0.00E+00	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03
Se-79	2.95E+05	7.17E-04	7.17E-04	7.16E-04	7.16E-04	7.16E-04	7.16E-04	7.16E-04	7.16E-04	7.15E-04	7.15E-04	7.15E-04	7.15E-04
Sm-147	1.06E+11	0.00E+00	9.82E-10	9.82E-10	9.82E-10	9.82E-10	9.82E-10	9.82E-10	9.82E-10	9.82E-10	9.82E-10	9.82E-10	9.82E-10
Sm-151	9.00E+01	6.57E+01	3.23E+01	1.50E+01	6.93E+00	3.21E+00	1.49E+00	6.88E-01	3.18E-01	1.47E-01	6.82E-02	3.16E-02	1.46E-02
Sn-121	3.09E-03	0.00E+00	3.73E-02	7.69E-03	1.58E-03	3.27E-04	6.74E-05	1.39E-05	2.87E-06	5.91E-07	1.22E-07	2.51E-08	5.18E-09
Sn-121m	4.39E+01	2.05E-01	4.80E-02	9.90E-03	2.04E-03	4.21E-04	8.68E-05	1.79E-05	3.69E-06	7.61E-07	1.57E-07	3.24E-08	6.67E-09
Sn-126	2.30E+05	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03	2.74E-03
Sr-90	2.88E+01	5.87E+03	6.41E+02	5.77E+01	5.19E+00	4.68E-01	4.21E-02	3.79E-03	3.41E-04	3.07E-05	2.77E-06	2.49E-07	2.24E-08
Tc-99	2.11E+05	2.57E-01	2.57E-01	2.57E-01	2.56E-01	2.56E-01	2.56E-01	2.56E-01	2.56E-01	2.56E-01	2.56E-01	2.56E-01	2.56E-01
Te-125m	1.57E-01	9.43E-02	2.11E-12	2.58E-23	3.16E-34	3.86E-45	4.72E-56	5.78E-67	7.06E-78	8.64E-89	1.06E-99	1.29E-110	1.58E-121
Th-227	5.12E-02	0.00E+00	4.70E-04	4.24E-04	4.22E-04	4.23E-04	4.23E-04	4.24E-04	4.24E-04	4.25E-04	4.25E-04	4.26E-04	4.27E-04
Th-228	1.91E+00	0.00E+00	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Th-229	7.34E+03	2.78E-04	6.93E-04	1.14E-03	1.58E-03	2.02E-03	2.45E-03	2.88E-03	3.31E-03	3.73E-03	4.15E-03	4.56E-03	4.97E-03
Th-230	7.54E+04	2.50E-04	3.33E-04	4.35E-04	5.44E-04	6.55E-04	7.67E-04	8.80E-04	9.92E-04	1.11E-03	1.22E-03	1.33E-03	1.44E-03
Th-231	2.91E-03	0.00E+00	6.90E-04	6.91E-04	6.92E-04	6.93E-04	6.94E-04	6.95E-04	6.96E-04	6.97E-04	6.98E-04	6.99E-04	7.00E-04
Th-232	1.41E+10	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03	1.56E-03
Th-234	6.60E-02	0.00E+00	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03

Index year:		2023	2115	2215	2315	2415	2515	2615	2715	2815	2915	3015	3115
Name	Half-Life (y)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)	Activity (µCi/g)
Tl-206	7.99E-06	0.00E+00	1.72E-11	3.64E-11	6.07E-11	9.02E-11	1.25E-10	1.64E-10	2.08E-10	2.57E-10	3.10E-10	3.67E-10	4.28E-10
Tl-207	9.08E-06	0.00E+00	4.76E-04	4.28E-04	4.27E-04	4.27E-04	4.28E-04	4.29E-04	4.29E-04	4.30E-04	4.30E-04	4.31E-04	4.31E-04
Tl-208	5.81E-06	0.00E+00	5.59E-04	5.59E-04	5.59E-04	5.59E-04	5.59E-04	5.59E-04	5.59E-04	5.59E-04	5.59E-04	5.59E-04	5.59E-04
Tl-209	4.11E-06	0.00E+00	1.45E-05	2.38E-05	3.31E-05	4.22E-05	5.13E-05	6.03E-05	6.92E-05	7.80E-05	8.67E-05	9.53E-05	1.04E-04
Tl-210	2.47E-06	0.00E+00	3.54E-09	6.80E-09	1.09E-08	1.58E-08	2.14E-08	2.78E-08	3.50E-08	4.29E-08	5.14E-08	6.06E-08	7.03E-08
U-233	1.59E+05	4.83E-02	4.82E-02	4.82E-02	4.82E-02	4.82E-02	4.82E-02	4.82E-02	4.81E-02	4.81E-02	4.81E-02	4.81E-02	4.81E-02
U-234	2.46E+05	8.80E-02	1.07E-01	1.16E-01	1.20E-01	1.22E-01	1.23E-01	1.23E-01	1.24E-01	1.24E-01	1.24E-01	1.24E-01	1.24E-01
U-235	7.04E+08	6.89E-04	6.90E-04	6.91E-04	6.92E-04	6.93E-04	6.94E-04	6.95E-04	6.96E-04	6.97E-04	6.98E-04	6.99E-04	7.00E-04
U-235m	4.95E-05	0.00E+00	1.02E+01	1.01E+01	1.01E+01	1.01E+01	1.01E+01	1.00E+01	1.00E+01	9.98E+00	9.95E+00	9.92E+00	9.89E+00
U-236	2.34E+07	1.39E-03	1.39E-03	1.40E-03	1.41E-03	1.42E-03	1.43E-03	1.44E-03	1.45E-03	1.46E-03	1.46E-03	1.47E-03	1.48E-03
U-237	1.85E-02	0.00E+00	9.84E-06	9.69E-08	1.92E-08	1.86E-08	1.87E-08	1.87E-08	1.87E-08	1.86E-08	1.86E-08	1.85E-08	1.84E-08
U-238	4.47E+09	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03	8.33E-03
U-240	1.61E-03	0.00E+00	3.56E-12	7.42E-12	1.13E-11	1.52E-11	1.90E-11	2.29E-11	2.67E-11	3.06E-11	3.45E-11	3.83E-11	4.22E-11
Y-90	7.32E-03	5.87E+03	6.41E+02	5.77E+01	5.19E+00	4.68E-01	4.21E-02	3.79E-03	3.41E-04	3.07E-05	2.77E-06	2.49E-07	2.24E-08
Zr-93	1.53E+06	5.95E-01	5.95E-01	5.95E-01	5.95E-01	5.95E-01	5.95E-01	5.95E-01	5.95E-01	5.95E-01	5.95E-01	5.95E-01	5.94E-01
Total	-	1.28E+04	1.51E+03	2.06E+02	6.65E+01	4.35E+01	3.61E+01	3.25E+01	3.05E+01	2.92E+01	2.83E+01	2.77E+01	2.71E+01
Reportable	-	6.54E+03	8.15E+02	1.33E+02	5.05E+01	3.27E+01	2.58E+01	2.24E+01	2.04E+01	1.91E+01	1.82E+01	1.76E+01	1.71E+01

Distribution:

William.bates@srnl.doe.gov Donna.Byrdy@srnl.doe.gov Brandi.clark@srnl.doe.gov georgiaanne.craig@srnl.doe.gov marion.cofer@srnl.doe.gov Vanessa.cofer@srnl.doe.gov alex.cozzi@srnl.doe.gov c.diprete@srnl.doe.gov David.diprete@srnl.doe.gov connie.herman@srnl.doe.gov sarah.hodges@srnl.doe.gov Charles.James@SRNL.DOE.GOV brady.lee@srnl.doe.gov kalyn.loudermilk@srnl.doe.gov Heather.Capogreco@srnl.doe.gov Joseph.Manna@srnl.doe.gov Gregg.Morgan@srnl.doe.gov frank.pennebaker@srnl.doe.gov Amy.Ramsey@srnl.doe.gov William.Ramsey@srnl.doe.gov eric.skidmore@srnl.doe.gov michael.stone@srnl.doe.gov boyd.Wiedenman@srnl.doe.gov Mary.whitehead@srnl.doe.gov Morgana.whiteside@srnl.doe.gov Jennifer.Wohlwend@srnl.doe.gov Records Administration (EDWS) jeffrey.crenshaw@srs.gov james.folk@srs.gov timothy.littleton@srs.gov tony.polk@srs.gov Anthony.Robinson@srs.gov Joseph.smith@srs.gov thomas.temple@srs.gov robert.hoeppel@srs.gov Thomas.Huff@srs.gov Vijay.Jain@srs.gov Crystal.Currie@srnl.doe.gov Azadeh.Samadi-Dezfouli@srs.gov jeff.ray@srs.gov Ryan.Lentsch@srs.gov matthew02.sims@srs.gov	terri.fellinger@srs.gov Joseph.fields@srs.gov bill.holtzscheiter@srs.gov Fabienne.Johnson@srnl.doe.gov Jeremiah.Ledbetter@srs.gov chris.martino@srnl.doe.gov aubrey.silker@srs.gov spencer.isom@srs.gov kirk.russell@srs.gov ryan.mcnew@srs.gov phoebe.fogelman@srs.gov matthew.siegfried@srnl.doe.gov donna.yarbrough@srs.gov shannon.strong@srs.gov melany.mini@srs.gov david.walker@srs.gov thuy.le@srs.gov stephanie.harrington@srs.gov danielle.turon@srs.gov jacob.montgomery@srs.gov alexander.choi@srnl.doe.gov	